

Received 28 March 2023, accepted 4 April 2023, date of publication 11 April 2023, date of current version 17 April 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3266291

RESEARCH ARTICLE

A Hybrid Method of Traffic Congestion Prediction and Control

TIANRUI ZHANG¹, JIANAN XU¹, SIRUI CONG¹, CHUANSHENG QU¹, AND WEIBO ZHAO²

¹School of Mechanical and Engineering, Shenyang University, Shenyang 110044, China

²School of International Education, Shenyang University, Shenyang 110044, China

Corresponding author: Jianan Xu (mikexu1226@163.com)

This work was supported in part by the China Central Guiding Local Science and Technology Development Fund Project “Application Research on Intelligent Collaborative Design and Control of Major Equipment Based on Cloud Service” under Grant 2021JH6/10500149, and in part by the Liaoning Provincial Natural Science Foundation Project “Research on Key Technologies of MRO Service for Complex Equipment Oriented to the Full Life Cycle” under Grant 20180551001.

ABSTRACT With the increasing complexity of urban transportation system, serious traffic congestion brings inconvenience to travel. It is also very difficult to predict and control traffic congestion. Therefore, this paper takes urban traffic condition and traffic congestion as the research object, and conducts in-depth research on traffic condition prediction model and traffic congestion control method. Firstly, a traffic state prediction method based on improved particle swarm optimization (IPSO) optimized radial basis function (RBF) and long and short term memory network (LSTM)/ support vector machine (SVM) feature fusion model was proposed for urban traffic state prediction. Experiments were carried out based on the regional traffic data of Shenyang Station, and compared with other algorithms, which verified the superiority of the feature fusion model based on IPSO-RBF and LSTM/SVM in this paper. Secondly, aiming at the problem of urban traffic congestion, a congestion section control method based on traffic allocation is proposed. Through VISSIM simulation and comparison with other traffic control schemes, the superiority of the congestion section optimization method proposed in this paper is verified.

INDEX TERMS Traffic condition prediction, feature fusion, traffic flow allocation, safety queuing factor.

I. INTRODUCTION

In the field of urban intelligent transportation, the judgment of traffic congestion is the main research direction. If the formation of current road congestion can be controlled in advance through the rules of traffic related information, economic losses and traffic accidents can be avoided and reduced. Traffic system is a huge space body with complex structure [1]. It is difficult for general methods to complete the processing of actual traffic data, and the obtained prediction results will always have a big deviation from the actual. Therefore, how to effectively find more accurate rules in the huge and complex traffic data is the main problem faced by the current intelligent transportation field. Similarly, road congestion control is also an important part of the field of

intelligent transportation. Combined with the characteristics of the current road structure, corresponding control means are provided in the research field to provide effective control decisions for the field of traffic management, so as to quickly dissipate the congested sections and avoid the continuous aggravation of traffic congestion.

Therefore, this paper uses the basic theory of traffic flow and traffic control theory to complete the prediction and control of traffic congestion, which is of great significance for guiding travel and traffic control.

In recent years, the continuous development of deep learning technology makes the prediction model based on neural network gradually introduced into the traffic situation prediction.

Luo and Zhao [2], [3], and [4] use Recurrent Neural Network (RNN) (especially Long Short Term Memory Network (LSTM) [5]) to predict traffic time series data in traffic flow.

The associate editor coordinating the review of this manuscript and approving it for publication was Maurice J. Khabbaz^{id}.

Guo et al. [6] proposed a deep space time 3D-CNN strategy to predict traffic flow data. Zhu et al. [7] built a network tracking model based on trajectory data before and after data on the basis of RNN and CNN. Lin et al. [8] proposed a model of generating adversarial network for traffic flow network prediction. Peng et al. [9] proposed a new type of dynamic graph recursion convolutional neural network, namely dynamic GR-CNN network, to represent the topological structure of urban traffic network and the temporal and spatial characteristics of traffic nodes. In addition, [10] and [11] also predicted the traffic conditions and achieved good results.

In the process of prediction, in order to improve the performance and accuracy of the prediction process, more scholars use mixed models or algorithms to carry out related research. The results show that in most cases, the mixed model is always better than the single model.

Guo et al. [12] applied three approaches of K-nearest neighbor, BPNN and support vector machine to traffic flow prediction. Ryu et al. [13] also adopted the K-nearest neighbor method with greedy search algorithm in short-term traffic prediction, and the above methods have good prediction accuracy in short-term traffic prediction. Kashi and Akbarzadeh [14] proposed a model based on the combination of wavelet transform and artificial neural network, in which wavelet transform is used for denoising and artificial neural network is used for predicting traffic. The results show that removal of noise can improve the accuracy of prediction obviously. Luo et al. [15] proposed a hybrid particle swarm optimization (PSO) algorithm combined with genetic algorithms to search the parameters of least squares support vector machines and thus predict short-term traffic flow. Feng et al. [16] proposed a hybrid algorithm of adaptive multi-core support vector machines based on temporal and spatial correlation. From the point of view of time and space. Duan et al. [17] and Dai et al. [18] have developed a unified temporal and spatial model to predict short-term traffic flows, all of which confirm the validity of the model. Duan et al. [19] developed a multimodal fractional Levy stable motion degradation model to predict the product technical life or remaining useful life (RUL) of equipment. The change point detection and clustering algorithm were used to identify the modes, and the validity of the prediction model was verified by an example. Song et al. [20] proposed a new heavy-tailed degradation model, in which the fractional Levy stable motion was taken as the diffusion term, a degradation model with power rate drift was established, and the differential iterative form of the fractional Levy stable motion degradation model was used to predict the remaining service life. Liu et al. [21] proposed a novel long-range-dependent (LRD) degradation model based on generalized Cauchy (GC) process. The dimensionless factor, principal component analysis (PCA) and iterative method were used to eliminate the interference of noise to the degraded data, and GC degradation model was used to predict the remaining useful life

(RUL) of rolling bearings and verify the effectiveness of GC degradation model.

In terms of traffic congestion control, Yang et al. [22] proposed a queue coordination strategy based on the “catch up” mechanism, and analyzed the impact of mixed traffic on the basic map, traffic oscillations, and traffic safety. The simulation results show that with the increase of market penetration rate (MPR) of autonomous vehicle, the road traffic capacity shows an increasing trend. Saleem et al. [23] proposed a fusion based VN intelligent traffic congestion control system (FITCCS-VN), which utilizes machine learning technology to collect traffic data and route traffic on available routes to alleviate traffic congestion in smart cities. The model improves traffic flow and reduces congestion. Atta et al. [24] discussed the application of RFID innovative technology in traffic congestion, and proposed an intelligent system that can maintain the dynamic time of traffic signals by sensing traffic density to minimize congestion.

Therefore, in the context of traffic congestion, there are few studies considering both traffic state prediction and control. With the increasing complexity of traffic problems, there are more and more research methods that combine intelligent algorithms with other related theories. Aiming at this key problem, this paper makes an in-depth study of traffic state prediction and congestion control, and proposes a traffic state prediction model based on the fusion of IPSO-RBF and LSTM/SVM features and a traffic distribution based congestion section control method. It is of great significance to establish the model of urban traffic congestion prediction and control. Therefore, this paper will study from the direction of traffic condition prediction and control.

II. TRAFFIC STATE PREDICTION MODEL BASED ON IPSO-RBF AND LSTM/SVM FEATURE FUSION

In this section, a traffic state prediction model based on IPSO-RBF and LSTM/SVM feature fusion is proposed based on the correlation between traffic running time and space and traffic flow parameters. Firstly, the basic ideas of traffic state prediction are analyzed, and the general process framework is established to preprocess traffic data. Then, IPSO-RBF and LSTM feature extraction models were constructed respectively, and the features were fused and SVM was used for traffic state prediction. Finally, the comparison of traffic state prediction algorithms is carried out and the results are deeply analyzed.

A. BASIC IDEA OF TRAFFIC STATE PREDICTION

In the process of traffic operation, many characteristics can be selected as the parameters of traffic state. In this paper, the parameters of traffic flow, density, speed and road occupancy are selected to comprehensively analyze the traffic state. Traffic road structure is a complex system, and is affected by various environmental factors, it is difficult to make a single analysis of different road forks and sections. Therefore, from

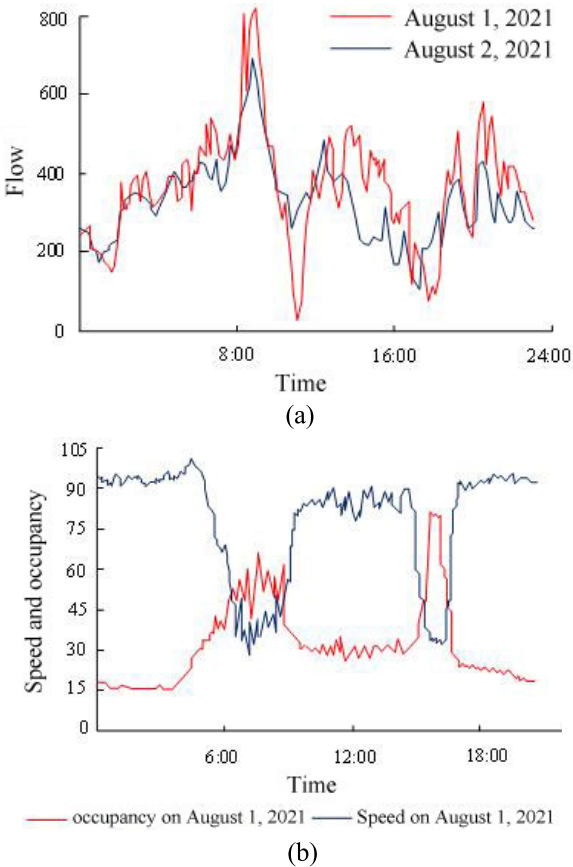


FIGURE 1. Change trend of traffic flow parameters. (a) Flow trend of two consecutive days; (b) Trend of speed and occupancy rate.

the macro perspective of traffic data analysis, the corresponding traffic operation rules can be obtained.

(1) Periodicity: Figure 1 (a) is shown that the traffic flow from August 1 to August 2, 2019 was detected at the same location on the traffic road, and there was a slight difference in the daily traffic flow, but the overall change trend was the same.

(2) Time correlation: Figure 1 (a) shows that the traffic flow is a continuous curve in a period of time, and the adjacent time nodes have strong correlation. Therefore, the research on traffic state prediction under normal operation can predict the following traffic conditions through historical traffic data.

(3) Spatial correlation: in the traffic road structure, all forks and sections are interrelated, and adjacent junctions or sections have strong correlation. Therefore, the characteristics analysis of flow and density of a single fork or road section can be used as a reference for the analysis of traffic status equivalence at other locations.

(4) Parameter correlation: It can be seen from Figure 1 (b) that with the increase of the change trend of the road occupancy rate, the vehicle driving speed also decreases, and when the road occupancy rate and the speed reach a moderate level, the traffic flow also increases. Therefore, there is a

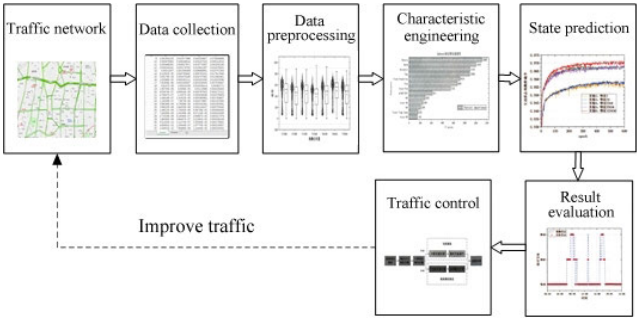


FIGURE 2. Traffic state prediction framework.

TABLE 1. Traffic state prediction framework.

Steps	Process
Step 1 Collect data	Traffic status information recorded by sensors
Step 2 Data preprocessing	Filter and repair the data
Step 3 Feature selection	Feature construction and feature selection
Step 4 Traffic condition prediction	Establish the traffic state prediction model
Step 5 Evaluation of prediction results	Traffic state classification and prediction
Step 6 Traffic control Applications	Get the traffic jam state

TABLE 2. Data acquisition tool.

Tool name	Version
Python	3.6.0
pandas	0.20.3
Numpy	1.16.2
requests	2.18.4
Arcgis	10.5

strong correlation between the parameters, which is of great significance to the overall traffic state analysis.

From the above analysis, it can be seen that although the traffic road structure itself is intricate and complex, it is not necessarily without rules to follow. In order to predict the traffic operation state more accurately, this paper comprehensively analyzes the traffic periodicity, time, space and parameter correlation through the macroscopic perspective of traffic operation characteristics, and uses the prediction model method of IPSO-RBF and LSTM/SVM feature fusion to complete the research of traffic state prediction.

In order to make better use of traffic prediction methods, this paper gives a general flow description of traffic state prediction work and puts forward the following traffic state prediction framework, as shown in Table 1 and Figure 2 below.

B. TRAFFIC DATA ACQUISITION AND PREPROCESSING

In this paper, part of the traffic data in Shenyang, Liaoning Province were collected for the experiment. The data comes from the open data platform of the Gaode Map. The data collection time granularity is 4 minutes, and a total

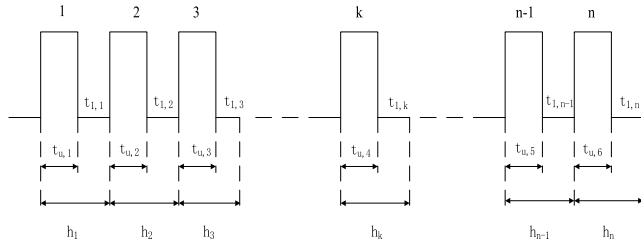


FIGURE 3. Principle of toroidal coil detector.

TABLE 3. Filtering rules for abnormal traffic data.

Data filtering rules	Abnormal judgment criterion
Rule 1	The speed, flow, or lane ratio of the vehicle is less than 0
Rule 2	The flow of vehicles is greater than the lane capacity
Rule 3	Lane occupancy is greater than 100%
Rule 4	Exceeding the maximum speed limit
Rule 5	Difference between the speed, flow and lane ratio of the vehicle
Rule 6	Only flow, lane ratio cannot be equal to 0 or only speed, lane ratio cannot be equal to 0

of 2 hours of data are obtained. The platform data can be obtained by using the Python crawler to connect with the official API. Table 2 shows the relevant software packages and version information required for data collection. The system acquired traffic data by setting relevant sensors in the key positions of the road, with more than 4460 monitoring points and more than 500 controllers, and the cumulative monitoring exceeded 30 kilometers. It is a common public data set in the field of traffic research. Taking the ring coil sensor as an example, when a vehicle is detected at the corresponding monitoring point, corresponding signals will be generated on the sensor, which will be output in the form of square waves and other information, such as vehicle flow, vehicle speed and occupancy rate, will be obtained from the sensor. The principle of ring coil detector is shown in Figure 3.

In the process of traffic data acquisition, a series of factors such as monitoring equipment in the complex environment will cause inevitable error effects. Error values or abnormal data will appear, which will affect the prediction results of the subsequent traffic prediction work. In order to avoid this situation, the collected raw data will be screened and processed in this paper. The main test period is the congestion peak, and abnormal data screening rules are shown in Table 3.

In this paper, abnormal data are identified, and the upper and lower limits of corresponding indexes are calculated, analyzed and compared by box diagram. The data beyond the upper and lower limits are regarded as abnormal values. Box

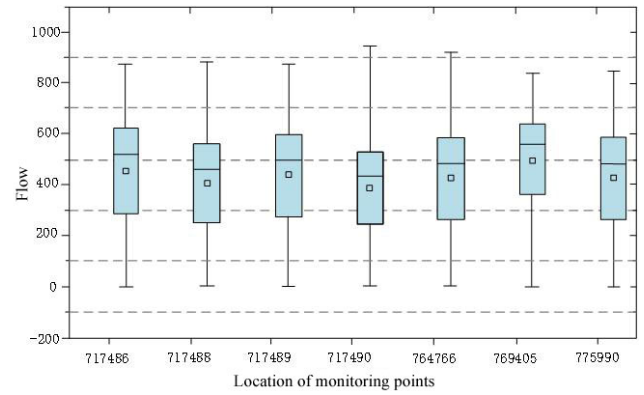


FIGURE 4. Box diagram of vehicle flow at each detection point.

patterns of vehicle flow at each monitoring point are shown in Figure 4.

After screening and processing the original data, the selected abnormal data will be repaired. The basic principle of data restoration is as follows:

$$x'_i = x_i d_i + \hat{x}_i (1 - d_i) \quad (1)$$

In the formula: x'_i —Data recovery value; x_i —Original data value; $\hat{x}_i$ —Estimate of the data; d_i —0 indicates that the data is abnormal. 1 indicates that the data is normal.

In this paper, the historical data, time series and spatial correlation abnormal data are repaired by the following methods.

Historical data restoration:

$$\hat{x}_i = x_{i-T} \quad (2)$$

Time series repair:

$$\hat{x}_i = \frac{\alpha_{i-1}x_{i-1} + \alpha_{i-2}x_{i-2} + \dots + \alpha_{i-k}x_{i-k}}{k} \quad (3)$$

Spatial correlation repair:

$$\hat{x}_i(j, m, n) = \gamma_0 + \gamma_1 x_i(m) + \gamma_2 x_i(n) \quad (4)$$

$$\hat{x}_i(j) = \text{median}[\hat{x}_i(j, m, n)] \quad (5)$$

In the formula: $\hat{x}_i$ —Data restoration value at time i ; x_{i-T} — $i - T$ Monitor the value at all times; k , α_i , γ_0 , γ_1 , γ_2 —Equation parameters; $\hat{x}_i(j, m, n)$ —Location (j, m, n) Data restoration value; $x_i(m)$, $x_i(n)$ —Monitoring values for position m and position n ; $\hat{x}_i(j)$ —Monitor j final repair value.

In this paper, the traffic state is classified and analyzed, and the traffic delay is divided into unimpeded, crowded and congested states. When the time of the traffic delay part is 0, the traffic state of this section is unimpeded. When the time of the traffic delay part is 0~2 minutes, the traffic state of this section is crowded. When the time of the traffic delay exceeds 2 minutes, the traffic state of this section is congested. By referring to the feature engineering method, each traffic parameter of the traffic delay part is set. The specific steps are shown in

TABLE 4. Feature engineering traffic parameters are set.

Step	Content
Feature establishment	Time characteristics: Traffic congestion is cyclical Density characteristics: Select vehicle speed and flow parameters to determine density characteristics
Feature selection	Number of balance control features
Feature scaling	Standardized treatment

Table 4.

$$x'_i = \frac{x_i - \bar{x}}{s} \quad (6)$$

s — Standard deviation; $\bar{x}$ —Mean.

C. TRAFFIC STATE PREDICTION BASED ON IPSO-RBF AND LSTM/SVM MODELS

1) THE IPSO-RBF SPATIOTEMPORAL FEATURE EXTRACTION MODEL IS ESTABLISHED

With the change of time deduction and uncertainty factors, the traffic operation will become unpredictable, so it is difficult to extract the spatial and temporal characteristics of the irregular traffic operation data with high accuracy through the traditional algorithm model. Using this characteristic of RBF neural network, it can effectively realize the nonlinear data processing of traffic running state and output the spatiotemporal characteristics of traffic. Firstly, the IPSO algorithm was combined with RBF neural network, and the IPSO algorithm was used to optimize the center vector, base width parameters and connection weights of RBF neural network. Then input the sample data to complete the training of RBF neural network. IPSO optimized the output model flow of RBF neural network, as shown in Figure 7.

Because the traditional PSO algorithm to optimize the output precision of RBF neural network mainly depends on the model of the center vector, the width of the vector and the connection weights of the algorithm performance, therefore, this article introduced IPSO algorithm, in the process of iteration, if the judgment algorithm have not reached a certain degree of convergence, removal of the optimal value corresponding to the particle, The other particles were processed by genetic algorithm crossover and mutation, and the other particles were randomly initialized for optimization solution verification. The improved particle swarm optimization algorithm is applied to RBF neural network to optimize the center vector, base width parameters and connection weights, which solves the problem that the parameters of conventional algorithms are easy to fall into local minima in determining RBF neural network, so that RBF neural network has higher extraction accuracy of traffic spatiotemporal features.

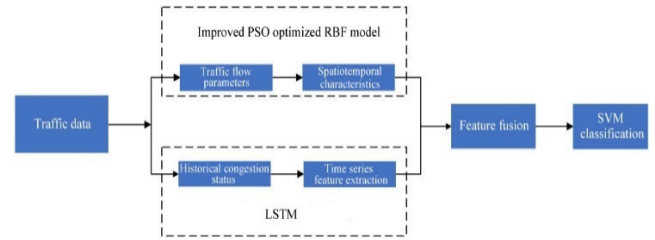


FIGURE 5. Feature extraction structure diagram of IPSO-RBF and LSTM.

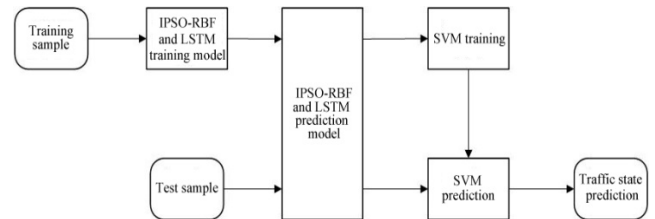


FIGURE 6. IPSO-RBF and LSTM/SVM model training process.

2) TEMPORAL AND SPATIAL FEATURE EXTRACTION OF TRAFFIC BASED ON IPSO-RBF AND LSTM

When analyzing the time-space characteristics of traffic operation, considering the correlation between traffic flow parameters and historical traffic status and time-space characteristics and time series characteristics, the upcoming traffic operation status can be predicted. In order to better solve the problem of spatial and temporal differences in traffic operation, this paper proposes a feature extraction model based on IPSO-RBF and LSTM, which respectively extracts the spatial and temporal features of traffic operation, fuses the extracted data, and improves the prediction results of traffic conditions. The IPSO-RBF and LSTM feature extraction structure is shown in Figure 5.

In this paper, the training flow chart of IPSO-RBF and LSTM models is shown in Figure 6. Firstly, some traffic flow training samples are extracted from the original traffic operation data as the training set of IPSO-RBF and LSTM models to train the models respectively. Then, the traffic flow test samples were extracted from the original traffic operation data as the test set of the trained model, and the features of the trained model were extracted. Finally, the feature extraction results of the training set and the test set are respectively trained and predicted by SVM, and the traffic running state is predicted and classified.

3) ESTABLISH TRAFFIC STATE PREDICTION MODEL BASED ON IPSO-RBF AND LSTM/SVM FEATURE FUSION

Traffic data is transformed into a two-dimensional eigenvector matrix by using temporal and spatial correlation. The time data monitored at monitoring point P is taken as the horizontal sequence of the characteristic matrix; According to the spatial position monitored by the monitoring point P , it is taken as the longitudinal sequence of the characteristic matrix, and x_{pt} is the traffic operation condition monitored by the monitoring

point p at time t , and A is the input matrix of the monitoring point:

$$A = \begin{pmatrix} x_{p-U,t-1} & x_{p-U,t-2} & \cdots & x_{p-U,t-T} \\ \vdots & \vdots & \ddots & \vdots \\ x_{p-1,t-1} & x_{p-1,t-2} & \cdots & x_{p-1,t-T} \\ x_{p,t-1} & x_{p,t-2} & \cdots & x_{p,t-T} \\ x_{p+1,t-1} & x_{p+1,t-2} & \cdots & x_{p+1,t-T} \\ \vdots & \vdots & \ddots & \vdots \\ x_{p+D,t-1} & x_{p+D,t-2} & \cdots & x_{p+D,t-T} \end{pmatrix} \quad (7)$$

In the formula: x_{pt} —Features monitored by monitoring point P at time t ; U —Number of upstream monitors; D —Number of downstream monitors; T —Length of time interval.

The IPSO optimized RBF model is used to input and output the structural mode of the sliding time window for samples. The horizontal axis direction of the sliding time window is set as the time series of traffic operation, and the vertical axis direction is set as the spatial layout position of traffic operation monitoring points. Each small rectangle represents the characteristic vector of traffic operation at different time nodes. Time $t+1$ is the sample tag dataset, time t is the detection set corresponding to the sliding time window, and there is a corresponding time interval between them. In this paper, the time interval is set to 15 minutes, so the sample tag on the horizontal time series is 15 minutes ahead of the sliding time window. IPSO optimizes the sliding time window of RBF model to construct the input and output sample mode, as shown in Figure 8.

Similarly, the traffic operation conditions monitored at monitoring point P have certain time series correlation, and the LSTM model is used to process the time-series features. The time interval of the LSTM model input is set to 15 minutes, at which time the LSTM and IPSO-RBF models output the traffic operation features of the same monitoring point. By applying the completed training IPSO-RBF and LSTM models to the traffic operation test set for spatio-temporal and temporal feature extraction, and combining the serial approach to convert two one-dimensional vectors into a new one-dimensional vector as the input value of the SVM. The output sample labels of SVM and IPSO-RBF and LSTM models are guaranteed to be consistent.

Traffic state prediction is mainly divided into three parts: traffic operation data preprocessing, IPSO-RBF and LSTM feature extraction model training and feature fusion SVM classification prediction. The traffic state prediction process of IPSO-RBF and LSTM/SVM feature fusion is shown in Figure 9.

The first part is to obtain traffic data, analyze the traffic data, first judge whether the data is lost, and if there is data loss, repair the data; Then, the complete data is judged to be abnormal. If the data is abnormal, it needs to be repaired, and the repaired data and complete data are regarded as reliable data; In the process of feature engineering analysis of reliable data, the feature engineering structure is determined, the data is denoised and normalized, and the sample output

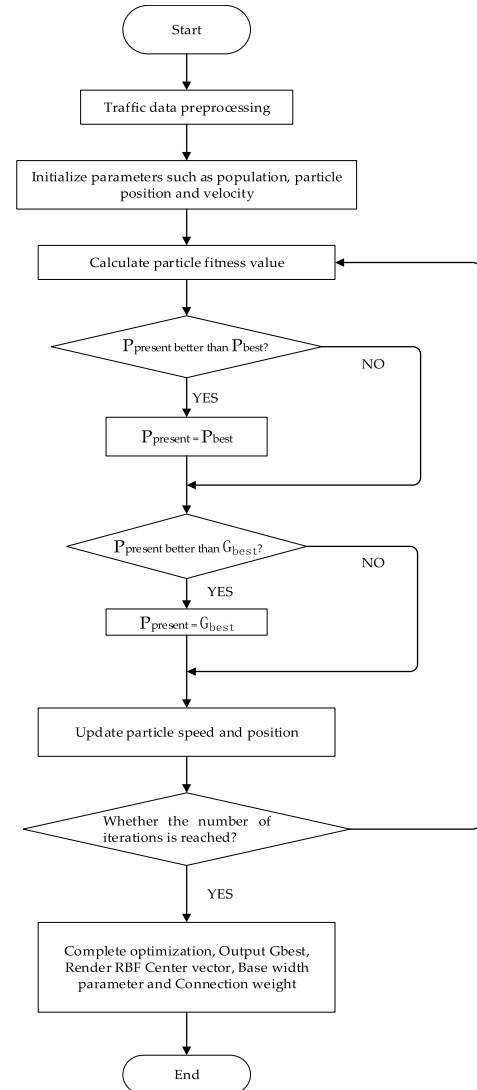


FIGURE 7. Flow chart of IPSO-RBF neural network model.

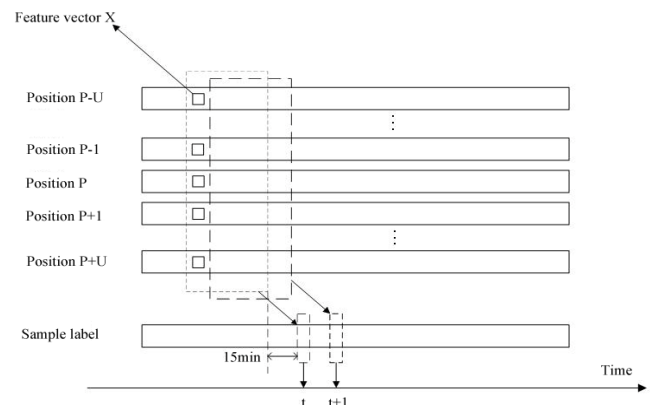


FIGURE 8. Sliding time window to construct input and output samples.

is constructed by time window sliding, and the data is input into the subsequent model.

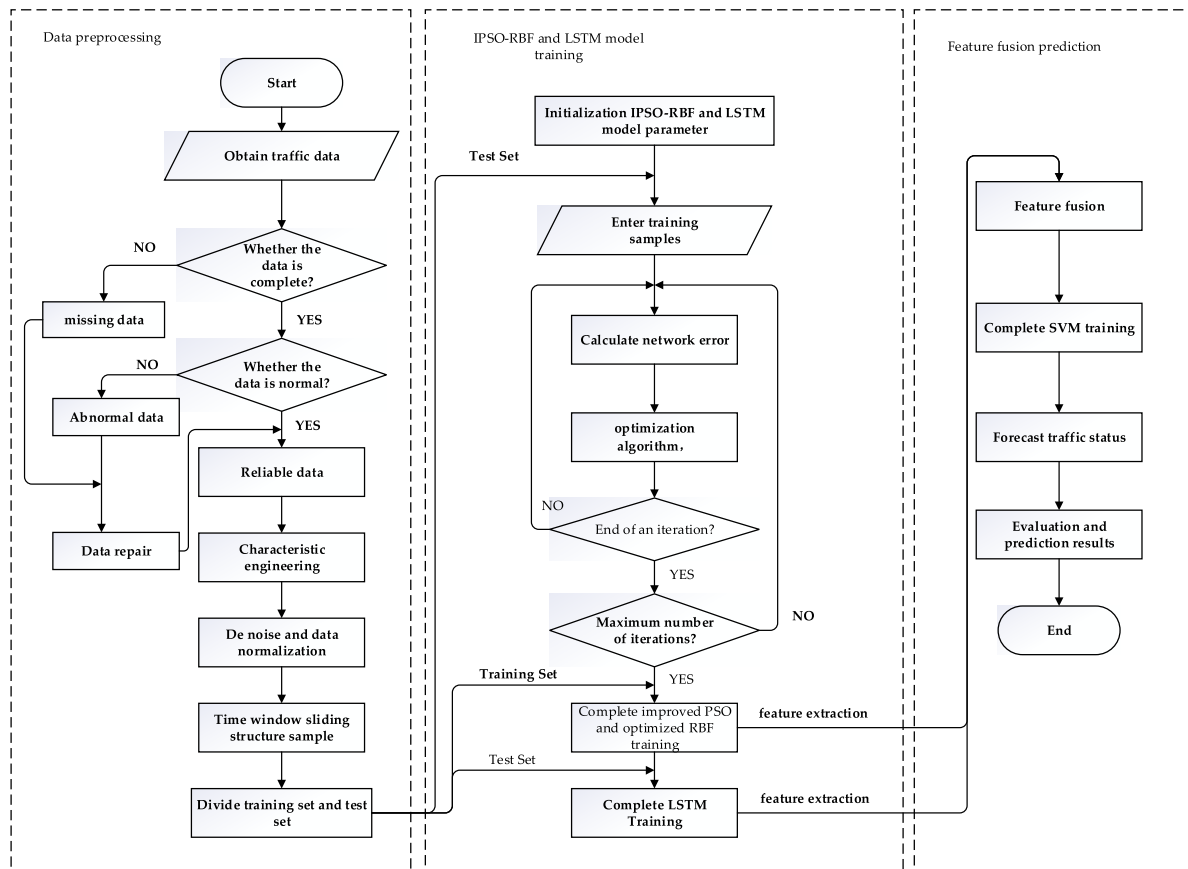


FIGURE 9. Flow chart of traffic state prediction based on feature fusion of IPSO-RBF and LSTM/SVM.

The second part: Construct IPSO-RBF and LSTM models, set the initial model parameters, and use the data and processed training samples as the input of the training model. The network error is further optimized through model training to calculate the network error, and then the algorithm is iterated to determine whether the iteration is over. If the maximum number of iterations is not over, the iteration experiment will continue until the IPSO-RBF and LSTM models are trained. The data and processed test sets are input into the trained model to complete the feature extraction of time, space and time sequence.

The third part: Conduct feature fusion on the extracted time-space and time series features, obtain feature fusion training set and test set, complete the training of SVM, input the test set into the SVM that has completed the training to make classified prediction of traffic operation status, further comprehensively evaluate the prediction results, and provide effective guidance for the actual traffic system.

D. CONGESTION CONTROL METHOD BASED ON TRAFFIC ASSIGNMENT

Traffic congestion usually occurs from one road section to the surrounding road sections. The main reason is that the queue length of vehicles is longer than the length of the

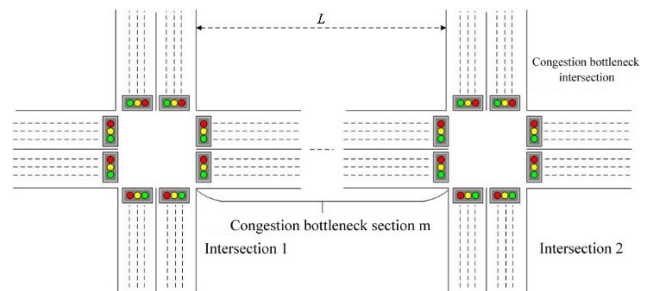


FIGURE 10. Basic structure of adjacent intersections and congested road sections.

road, which leads to traffic congestion. The congested section is marked as section m , the length of congested section is L , intersection 2 is a congested intersection, and the basic structure of adjacent intersections and congested sections is shown in Figure 10.

When the traffic operation is congested, the capacity of the intersection at the congested section is usually adjusted to the maximum capacity, and the cycle of vehicle traffic is set to the maximum cycle. When the queue length of vehicles is too long, in order to reduce the queue length of congested sections to the normal queue length L_0 within time T , the

output capacity difference ΔS_1 of traffic congested sections is:

$$\Delta S_1 = \frac{L_{\max} - L_0}{T \cdot l} \varphi \quad (8)$$

$$S_{out,m} = \sum_{h \in L_{out}^m} (\lambda_{m,h} \cdot Q_{m,h}) \quad (9)$$

In the formula: $L_{\max}$ —Maximum queue length of vehicles; l —Vehicle spacing; φ —Number of roads; $S_{out,m}$ —Vehicle output degree of track m ; $S_{in,m}$ —Vehicle input degree of track m ; $Q_{m,h}$ —Vehicle output rate of each section; $Q_{z,m}$ —Vehicle input rate of each section; $\lambda_{m,h}$ —Green ratio; $\lambda_{z,m}$ —Minimum green ratio.

$$S_{in,m} = \sum_{h \in L_{in}^m} \min(q_{z,m}, \lambda_{z,m} \cdot Q_{z,m}) \quad (10)$$

In the formula: L_{out}^m —output set of all vehicles in track m ; L_{in}^m —input set of all vehicles in track m .

In the road section under conventional control, there is no queuing situation caused by vehicle congestion in the upstream road section, and the poor upstream and downstream input and output capacity of the traffic congested road section S_2 is:

$$\begin{aligned} \Delta S_2 &= S_{out,m} - S_{in,m} \\ &= \sum_{h \in L_{out}^m} (\lambda_{m,h} \cdot Q_{m,h}) - \sum_{z \in L_{in}^m} \min(q_{z,m}, \lambda_{z,m} \cdot Q_{z,m}) \end{aligned} \quad (11)$$

When there is a queue situation caused by traffic congestion upstream:

$$\begin{aligned} \Delta S_2 &= S_{out,m} - S_{in,m} = \sum_{h \in L_{out}^m} (\lambda_{m,h} \cdot Q_{m,h}) \\ &- \sum_{z \in L_{in}^m} (\lambda_{z,m} \cdot Q_{z,m}) \end{aligned} \quad (12)$$

Therefore, in case of traffic jam, the traffic capacity ΔS of this section shall be increased in time as follows:

$$\Delta S = \Delta S_1 - \Delta S_2 \quad (13)$$

According to the traffic road to build the corresponding structure of road congestion, the circular connection between the regional said road intersection, the connection between the circular area part of traffic, said traffic congestion through color, green represents the road traffic status to smooth state, red for road traffic state to congestion state. In this paper, the red part is defined as the first-level congested road section, which will be downgraded and classified as the color changes to the green trend. The higher the correlation degree of traffic congestion is, the more obvious the regulation effect will be. The intention of the congestion degree of the congested section and its adjacent sections is shown in Figure 11.

The traffic operation of adjacent sections of traffic congested sections is analyzed, and the remaining capacity of this section is used to measure the traffic flow that can be borne

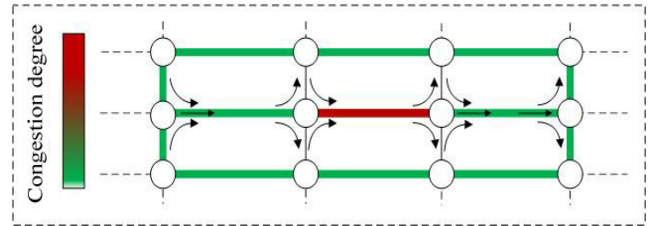


FIGURE 11. Congestion degree intention of congested road section and adjacent road section.

by adjacent sections of traffic congested sections, as shown in Formula (14).

$$V_i = \frac{\alpha L_i - L_i \varphi_i}{l} \quad (14)$$

In the formula: α —Road safety queuing factor; $\bar{L}_i$ —length of the road; L_i —Current queue length; l —Front spacing; φ_i —Number of lanes.

The congested road section produces different delay degrees under different vehicle queuing conditions, which can be divided into three kinds of delay degrees according to the vehicle queuing conditions, as shown in Figure 12.

Where, v represents the arrival speed of vehicles, c represents the traffic capacity, and s represents the saturation rate. It can be seen from Figure 12 (a) that the yellow part in the figure indicates that when the arrival speed of the vehicle is consistent with the traffic capacity or does not exceed the traffic capacity, the vehicle is in the normal queue delay; It can be seen from Figure 12 (b) that the red part in the figure indicates that when the arrival speed of vehicles exceeds the traffic capacity in a short period of time, short-term queuing congestion will occur to the traffic operation, at which time the vehicles are in a general queuing delay; It can be seen from Figure 12 (c) that the red part in the figure indicates that when the arrival speed of vehicles far exceeds the traffic capacity, there will be a long period of congestion in this section of the road. The road is in a high load operation state, and the vehicles cannot be effectively evacuated in a short time. At this time, the vehicles are in a serious degree of delay.

Caused by considering the uncertainty factors are part of the road, according to the vehicle queue length cannot be accurately forecast to late vehicles that have accumulated increase, so the choice of reasonable safety coefficient of line, to control the actual future sections can be effectively improved, as shown in Table 5, based on the residual capacity of road section of potential road congestion control processing.

First, determine the congested road section and the adjacent congested road section, then predict the future traffic state of the adjacent congested road section, obtain the corresponding vehicle queue coefficient according to the future traffic state and calculate the residual capacity of the road section, further judge whether the road section is controllable or potential congested road section, and implement the corresponding allocation scheme to adjust it reasonably.

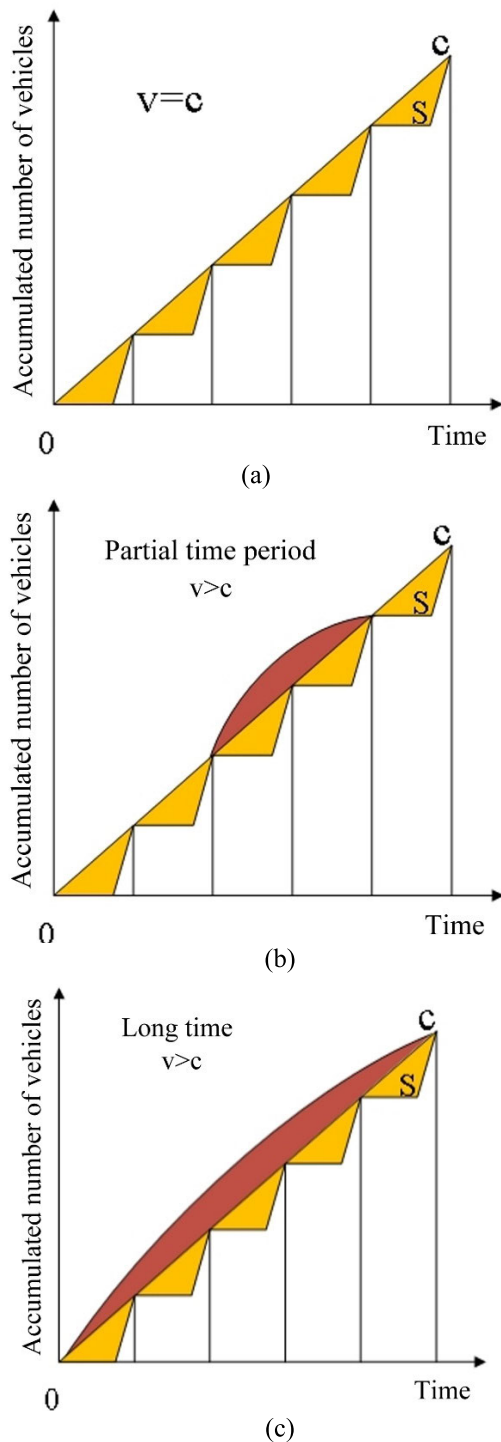


FIGURE 12. Three kinds of delays in different queuing situations.

E. DESIGN PRINCIPLE OF TRAFFIC CONGESTION PREDICTION AND CONTROL BASED ON DIGITAL TWIN TECHNOLOGY

The vehicle and road related elements in the traffic operation environment are moved from the physical space to the digital space, so as to achieve the intercommunication effect between the real traffic scene and the virtual scene. By using the

TABLE 5. Congestion potential road section control processing based on remaining capacity of road section.

Step	Contents
	Determine the bottleneck section R_0 and the bottleneck-related
1:	section R_i through the trigger threshold L_{begin} of the bottleneck section
2:	For each R_i do
3:	Predict the congestion state S_i
4:	after the time T of the bottleneck-related link
5:	If S_i is the congestion state
6:	$\alpha_i = \alpha$ // α is the safe queuing coefficient corresponding to the congestion state
7:	Else if S_i is the crowded state
8:	$\alpha_i = b$ // b is the safe queuing coefficient corresponding to the crowded state
9:	Else
10:	$\alpha_i = c$ // c is the safe queuing coefficient corresponding to the unblocked state
11:	$V_i = \alpha_i \left(\bar{L}_i - L_i \right) \phi_i / l$
12:	If $V_i > 0$
13:	Road R_i is a controllable section and needs to be adjusted for the timing plan
14:	else
15:	Road R_i is a potential bottleneck section, keeping the original timing plan
	End for

corresponding equipment to collect traffic data, the traffic running condition of the real scene can be timely reflected to the twin scene. At the same time, the traffic state of the real scene can be observed and predicted through virtual simulation in the twin scene, and the feedback and control of the traffic congestion of the real scene can be achieved in time. The combination of digital twin technology in the intelligent transportation system to realize the blending of virtual reality and iterative interaction of virtual control and real can effectively solve the complex traffic problems and promote the breakthrough of the research and development of intelligent transportation technology. Figure 13 shows an example of digital twin construction analysis for a traffic operation scenario.

The process of building the digital twin model of traffic congestion prediction and control is mainly divided into two

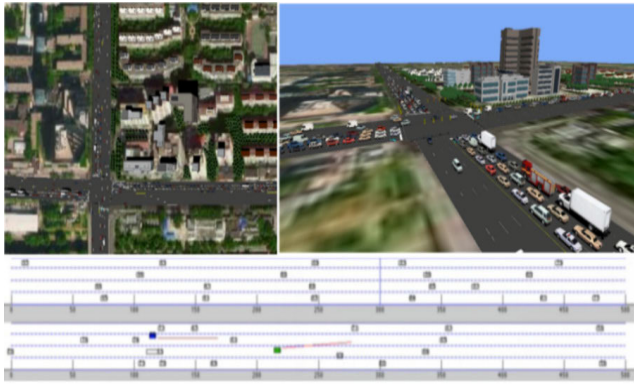


FIGURE 13. An example of digital twin construction analysis in a traffic scene.

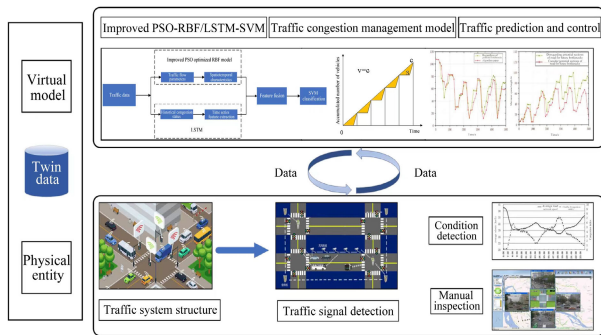


FIGURE 14. Virtual and real mapping framework for traffic congestion prediction and control.

important parts, namely physical traffic space and virtual traffic space; However, due to the digital characteristics of virtual space and the large differences between the collected corresponding space environment, road length, time nodes, vehicle flow and information transmission process and the physical traffic space, it is necessary to conduct a comprehensive information difference analysis on the research object before establishing the virtual reality mapping, so as to establish the traffic twin space more effectively and accurately.

In the actual traffic physical space operation process, the environment is complex, and the digital twin will have differences in virtual-real mapping. Therefore, it is necessary to accurately map the traffic resource information in virtual-real space to establish the virtual-real mapping framework for traffic congestion prediction and control, as shown in Figure 14.

On the basis of traffic congestion status data, on the one hand, it is necessary to establish various traffic signals, statuses, and manually detect congestion status based on the traffic system structure; on the other hand, it is necessary to obtain the corresponding congestion evaluation data, congestion prediction data, status and parameter data and other status information in real time according to the correlation between virtual and real space information. The collected

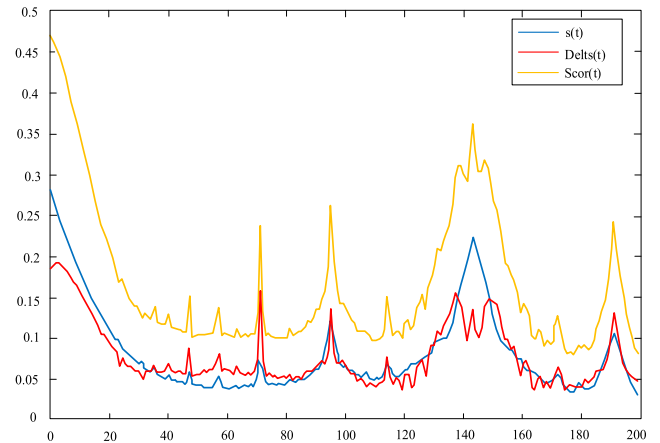


FIGURE 15. Phase space reconstruction diagram.

information is mapped to the virtual environment to achieve the combination of virtual and real.

First of all, according to the operation of the road traffic further analysis of city road structure characteristics, according to different characteristics of corresponding to the road traffic state data acquisition, including a variety of traffic flow and density, vehicle speed, the relevant data such as the length of the road, build into traffic on twin database. According to the traffic history data in real scenes collected by the traffic twin database, the advanced traffic state data is preprocessed and screened. Then the IPSO-RBF model and LSTM network are used to extract and fuse the processed data respectively, and the traffic state results are classified and predicted by SVM model. Finally, according to the introduction of the algorithm to deal with the congestion and potential sections, the traffic congestion flow distribution in the simulation software VISSIM system simulation experiment, and get an effective solution.

III. SIMULATION EXPERIMENT AND RESULT ANALYSIS

A. EXPERIMENTAL DESIGN

The traffic operation state itself has the correlation between time series and spatial characteristics, which is a complex stochastic nonlinear relationship. The experiment of RBF network and phase space reconstruction shows that when the number of RBF input layer nodes and the embedding dimension of phase space reconstruction are set to the same number m , the experimental results are better predicted. Therefore, the collected traffic flow time series features are reconstructed by C-C, and the reconstructed sequence is $X = [x_i, x_{i+\tau}, \dots, x_{i+(m-1)\tau}]^T$, $i = 1, 2, \dots, M$, m is the embedding dimension, τ is the delay time, and M is the number of phase points. Obtained through experiment $\tau = 31$, $\tau_w = 178$, $m \approx 7$ is finally calculated. The calculated embedding dimension is 7, so the number of nodes in the input layer of the neural network is set as 7. The relationship between multi-dimensional traffic data parameters is shown in Figure 15.

In this paper, IPSO algorithm is used to determine the hidden layer center vector, base width parameter and connection

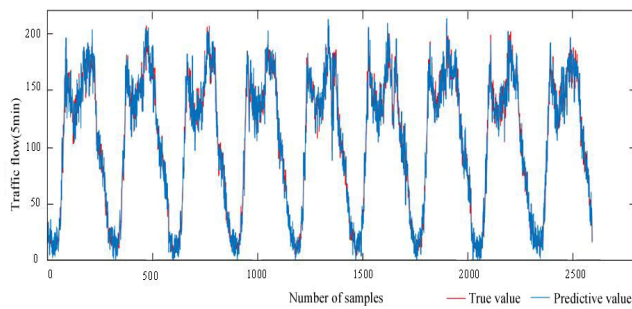


FIGURE 16. Elman neural network model fitting value.

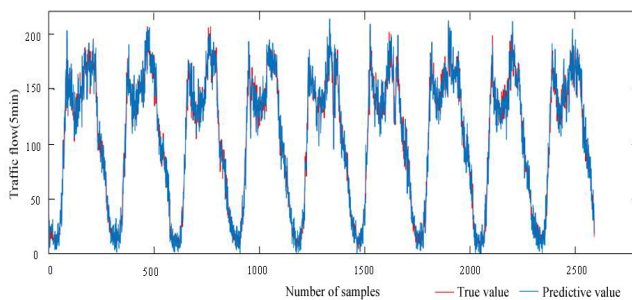


FIGURE 17. Fitted value of BP neural network model.

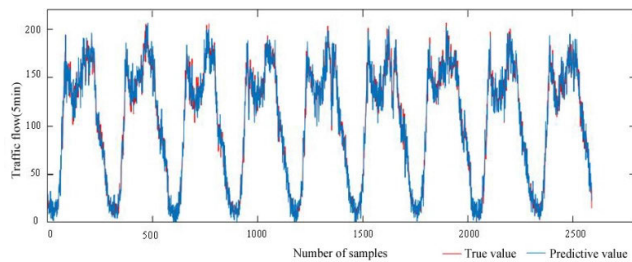


FIGURE 18. Fitted value of RBF and LSTM/SVM prediction model.

weight of RBF neural network. After many experiments, the base width parameter of RBF neural network is finally determined to be 11; The center and output weights of IPSO obtained by iterative calculation are used as the center and connection weights of neurons in each hidden layer; The output value is set as 1, so the number of nodes in the output layer of the neural network is 1. Therefore, the structure of RBF neural network is 7-11-1, and the number of model training is 5000.

In order to show that the traffic prediction state model with IPSO-RBF and LSTM/SVM feature fusion is more suitable to solve the traffic state prediction problem than other models, this paper uses Elman neural network, BP neural network and the traffic prediction method with RBF and LSTM/SVM feature fusion without optimization as the experimental comparison model. The fitting values of neural network model and prediction model are shown in Figure 16 to Figure 19. The training and prediction results of all the above models are shown in Figure 20, Figure 21.

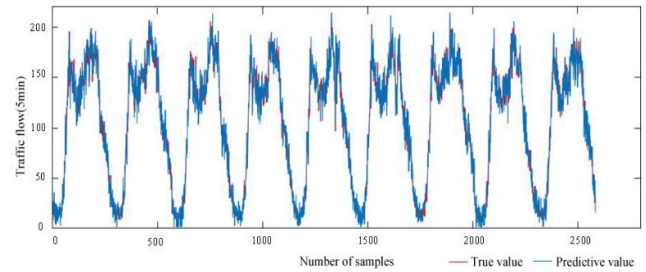


FIGURE 19. Fitted value of IPSO-RBF and LSTM/SVM prediction model.

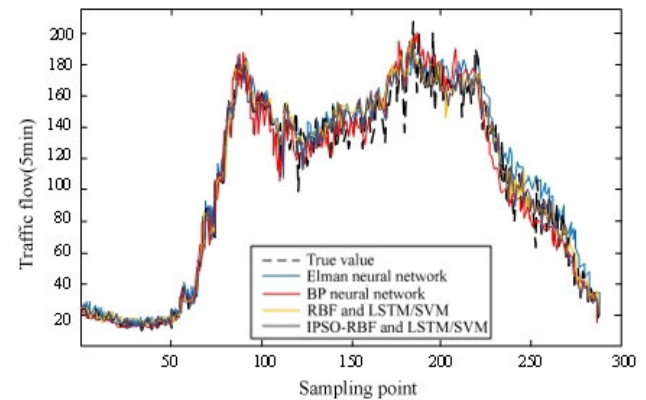


FIGURE 20. Comparison of experimental results of four models.

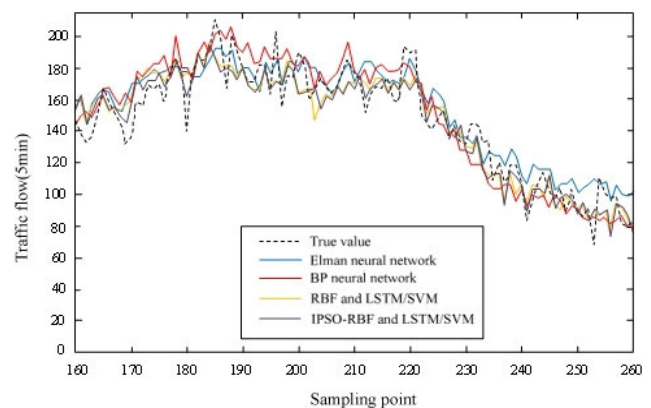


FIGURE 21. A partial comparison diagram of the experimental results of the four models.

It can be seen from Figure 19 that the trend of the curve of the predicted traffic flow of the four models after simulation training is roughly the same as that of the actual traffic flow, which shows that the predicted results of the four models after training are good. At the same time, it can be seen that the fitting values of the unoptimized RBF and LSTM/SVM prediction models are also closer to the actual values than the Elman neural network and BP neural network when the traffic flow is in the low valley or peak, and the traffic state prediction model with IPSO-RBF and LSTM/SVM features fusion has a better prediction effect.

It can be seen from Figures 20 and 21 that compared with Elman neural network and BP neural network, RBF and LSTM/SVM prediction models have the highest fitting degree between the predicted value of traffic flow and the actual value curve, indicating that RBF and LSTM/SVM prediction models are suitable for dealing with traffic state prediction problems with high complexity and strong nonlinearity; In addition, it can be seen that after the optimization of RBF neural network parameters through IPSO, IPSO-RBF and LSTM respectively extract and process the temporal and spatial features of the original traffic data, which improves the quality of traffic data. Finally, SVM is used for feature fusion to further improve the prediction accuracy.

In order to express the prediction result of the model intuitively, the corresponding evaluation index is used to evaluate the prediction result of the model comprehensively. The three evaluation indicators are defined as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - y'_i)^2} \quad (15)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n |(y_i - y'_i) / y_i| \times 100\% \quad (16)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y'_i| \quad (17)$$

$$AIC = 2k - 2 \ln(L) \quad (18)$$

$$BIC = k \ln(n) - 2 \ln(L) \quad (19)$$

In the formula: y_i —Actual traffic volume; y'_i —Model predicts traffic flow; n —Sample size; k —the number of estimated model parameters; L —the maximum value of the likelihood function for the model.

The prediction results of each model were evaluated quantitatively with root mean square error ($RMSE$), mean relative error ($MAPE$) and mean absolute error (MAE). The $RMSE$ value reflects the degree of error between the model predicted data results and the actual value. The smaller the $RMSE$ value is, the lower the degree of error is. The $MAPE$ value reflects the deviation degree between the predicted data results of the model and the actual value. The smaller the MAE value is, the lower the deviation degree of the model is, and the model has a good prediction effect. The smaller the values of AIC and BIC , the better the model. In order to further illustrate the effectiveness of the model presented in this paper, a comparative analysis of the errors of various prediction models is carried out by using quantitative analysis methods. The comparative analysis of the errors of the four prediction models is shown in Table 6.

It can be seen from Table 6 that the $RMSE$ value of the traffic state prediction model with IPSO-RBF and LSTM/SVM feature fusion decreases from 14.6219 of Elman neural network to 12.1543, the $MAPE$ value decreases from 14.3612 to 12.5234, and the MAE value decreases from 12.0214 to 10.0002, which are the lowest values of the four prediction models. The values of AIC and BIC are also the minimum

TABLE 6. Comparison and analysis of errors of four forecasting models.

Model evaluation	$RMSE$	$MAPE$	MAE	AIC	BIC
Elman neural network model	14.6219	14.3612	12.0214	-6.730	-6.772
BP neural network model	14.0567	14.0621	11.2804	-6.572	-6.858
RBF and LSTM/SVM prediction model	12.4201	12.8035	10.0025	-6.077	-7.353
IPSO-RBF and LSTM/SVM feature fusion traffic state prediction model	12.1543	12.5234	10.0002	-5.991	-7.511

values in each model. In addition, it can be seen that the prediction errors of RBF and LSTM/SVM prediction models are the lowest compared with Elman and BP neural networks, indicating that RBF and LSTM/SVM models are suitable for solving traffic prediction problems. It can be further seen that IPSO-RBF and LSTM/SVM feature fusion traffic state prediction models have indeed improved the prediction accuracy compared with RBF and LSTM/SVM prediction models, indicating that feature extraction of original traffic data improves the efficient use of traffic data and reduces the prediction error. At the same time, it also shows that the initial settings of the width parameter of the basis function, the center vector and the connection weight value of the traffic state prediction model with IPSO-RBF and LSTM/SVM feature fusion have a great impact on the quality of the prediction results in the process of predicting the traffic state. Improper selection of the parameters will reduce the prediction accuracy of the model on the traffic state.

B. EXPERIMENTAL ANALYSIS

In this paper, Dell Intel (R) Core (TM) i5-5200U CPU@2.20GHz, 2014 PTV AG VISSIM traffic simulation software is used to conduct experiments. VISSIM simulation software can simulate and set up real traffic scenes, and can establish complex traffic road structure. It can solve the complex problems of road architecture, and has the ability to reflect and simulate the actual traffic state efficiently. It can feedback the traffic running state better, and can be used in traffic simulation experiments. This paper takes the experimental path shown in Figure 22 as an example.

In Figure 22, $i + 1$ is a congested intersection, Section $L_{i+1,3}$ is a congested section, and a is a queue monitor. Let $L = \{L_j | j = 1, 2, \dots, I\}$ be the set of all intersections; j is the road number; I is the total number of intersections. In the simulation environment, each section is set to 3 roads.

In order to comprehensively evaluate the control effect of this paper on traffic congestion sections, the queue length, delay degree and stopping times of vehicles were analyzed.

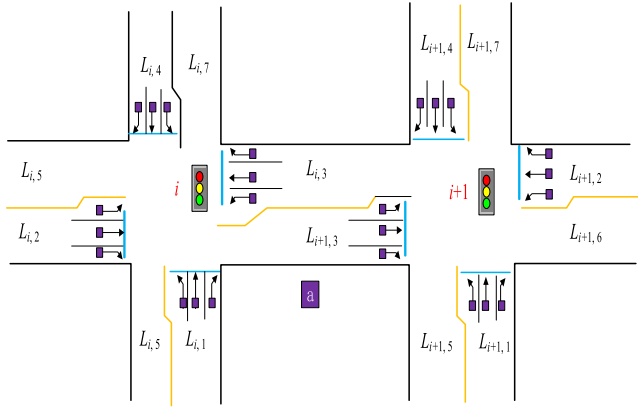


FIGURE 22. Simulation modeling experiment based on VISSIM.

(1) Vehicle delay degree:

$$d = d_{\text{balance}} + d_{\text{random}} + d_{\text{addition}} \quad (20)$$

(2) Vehicle queue length:

$$L = L_{\text{original}} + L_{\text{increase}} \quad (21)$$

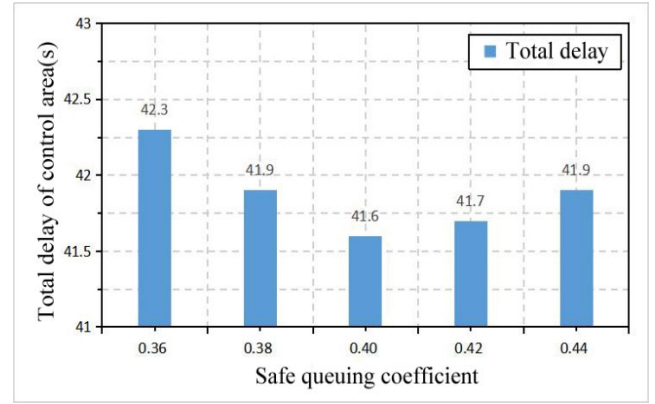
(3) Number of vehicle stops:

When the vehicle arrives at the intersection and waits to pass, the process from stationary to leaving is recorded as stopping once.

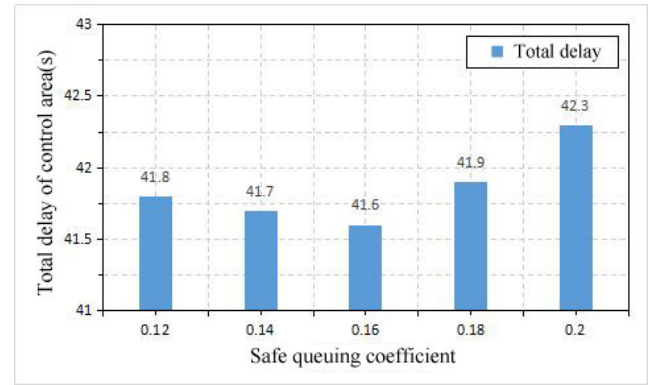
In the formula: d —Total delay; d_{balance} —Equilibrium delay degree; d_{random} —Uneven delay degree; d_{addition} —Plus degree of delay; L —Total vehicle queue length; L_{original} —Original accumulated length; L_{increase} —Current vehicle queue length.

The VISSIM simulation software is used to compare the congestion control method proposed in this paper with other methods. Set constraints on congestion control. When the vehicle queue length is greater than 0.5 times the length of the road section, the traffic state will enter the congestion state; When the length of the vehicle queue is less than 0.25 times the length of the road section, the road control decision will be ended. The disappearance time of traffic congestion is set as 600s, and the experimental results are analyzed as follows. The safety queuing coefficient is used to test the potential congested road sections, and the selection and analysis of the safety queuing coefficient of the road sections under various traffic conditions are shown in Figure 23.

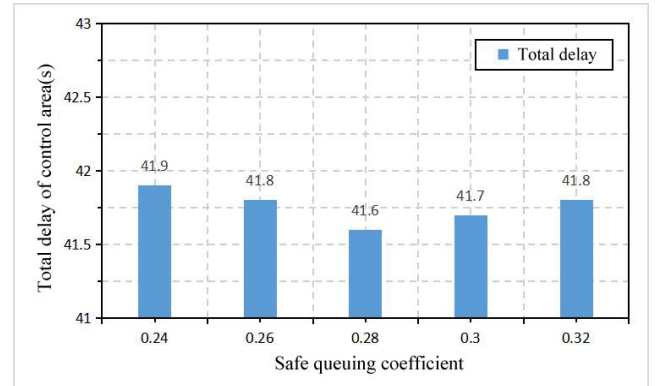
As shown in Figure 23 (a), when the road safety queuing coefficient is 0.4, the vehicle delay in the unblocked state is 41.6s. At this time, the vehicle delay is the lowest, indicating that the remaining capacity of the road in this state is large, and adjacent congestion sections can be adjusted; As shown in Figure 23 (b), when the road safety queuing coefficient is 0.16, the vehicle delay in the congestion state is 41.6s, at which time the vehicle delay is the lowest, indicating that the remaining capacity of the road is seriously insufficient, so it is necessary to control the traffic flow of this section and allocate the traffic flow to adjacent sections; As shown in Figure 23 (c), when the road safety queuing coefficient



(a)



(b)



(c)

FIGURE 23. The degree of delay in the safety queuing coefficient under the unblocked state.

is 0.28, the degree of vehicle delay in the congested state is also 41.6s, which means that the residual capacity of the road is average, which can ensure the input of traffic flow. At the same time, it is necessary to adjust the corresponding traffic flow to adjacent sections to reduce the potential traffic congestion pressure.

As shown in Figure 24 (a), the control effects in the two cases are roughly similar, but without considering the potential road section, too much traffic flow will be allocated to the road section, and the congestion speed of the road section will gradually increase. As shown in Figure 24 (b), without considering the potential road section, the growth rate of

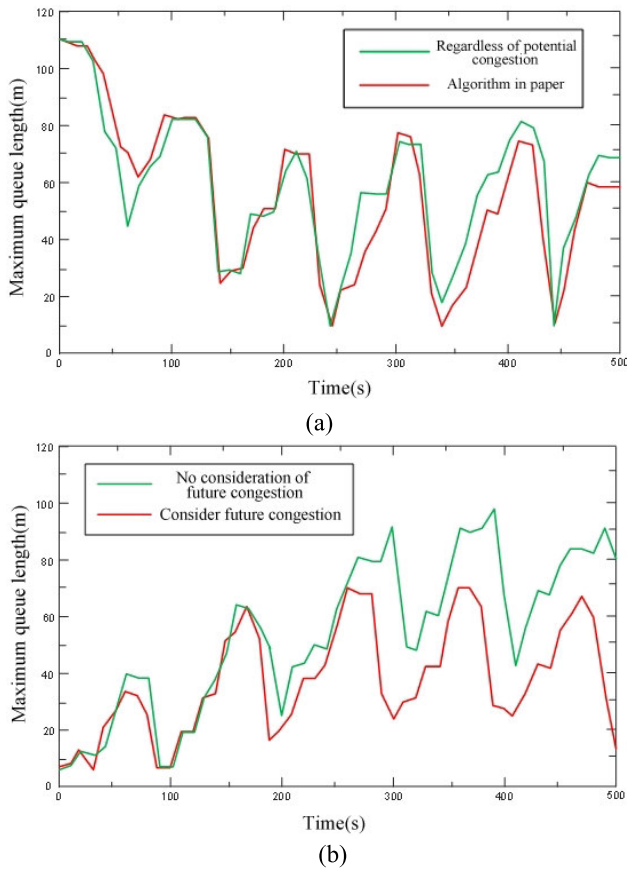


FIGURE 24. Comparison of the maximum queue length of potential congested road sections.

vehicle queue length in this section continues to accelerate, resulting in the formation of road congestion in advance and spread to adjacent sections, resulting in the acceleration of congestion in adjacent sections and aggravation of road congestion.

The traffic congestion section is generated when a large number of vehicles enter the section during the traffic peak period. It is difficult to control the traffic congestion state during the peak period only considering the traffic flow and road residual capacity when the peak period is not reached. Therefore, it is necessary to conduct a comprehensive analysis of the potential congestion sections. Therefore, this paper proposes a congestion section control method based on flow distribution to consider the potential congestion sections, keep the traffic in a smooth state.

C. ALGORITHM COMPARISON EXPERIMENT

In order to verify the control effect of this algorithm, this section will use other algorithms to conduct comparative experiments with this algorithm, mainly including the following four control algorithms. The control effects of the four congestion control algorithms are shown in Figure 25 to Figure 28.

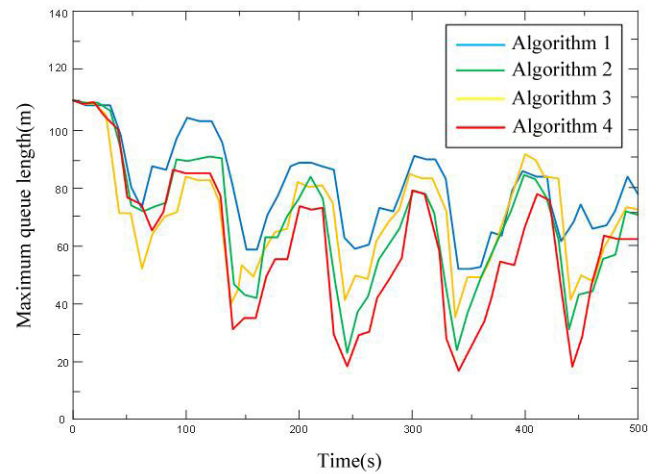


FIGURE 25. Change in maximum queue length.

Algorithm 1: Implement congestion control method for key road sections, and provide weight distribution control of traffic flow for key road sections.

Algorithm 2: control the traffic capacity of the current road, and provide the traffic capacity of the current section for traffic flow allocation control.

Algorithm 3: Provide the upstream and downstream traffic flow to limit the traffic capacity of the intersection according to the maximum traffic flow limit at the upstream and downstream of the congested road section.

Algorithm 4: This method adopts the vehicle safety queuing coefficient and road residual capacity to reasonably allocate the traffic flow on the congested road sections under the current and future potential conditions.

The algorithm in this paper has achieved the optimal control effect in the four evaluation aspects of L_{max} , L_{ave} , D and S . Compared with other algorithms, the main advantages of this algorithm are as follows:

- (1) Adjust the traffic capacity of the current traffic road and the vehicle flow carrying capacity;
- (2) Considering the safety queuing coefficient of vehicles in the case of potential congested road sections in the future, reasonably use the road occupation in advance;
- (3) The residual capacity of the traffic road is used to reasonably allocate the traffic flow, so that the road congestion can be quickly dissipated even in the traffic congestion state, effectively reducing and preventing the aggravation of the congestion.

The comparison results of various evaluation indicators of four traffic congestion control algorithms are shown in Table 7.

We consider the future traffic congestion, and divides the experiment into two parts: the current congested road section and the control area road section under the future traffic congestion. The results of the algorithm in this paper are the lowest in all evaluation indicators, which shows that the

TABLE 7. Comparison results of various evaluation indexes of different congestion control algorithms.

Evaluation index		Algorithm m 1	Algorithm m 2	Algorithm m 3	Algorithm m 4
Traffic jam	$L_{\max} / m$	83.42	80.55	87.71	73.24
	L_{ave} / m	53.63	49.92	47.73	47.11
	D / s	49.4	45.5	46.3	42.3
	$S / times$	1.13	1.22	1.11	0.96
	$L_{\max} / m$	83.42	80.55	87.71	73.24
Control region	L_{ave} / m	47.39	45.32	44.68	44.37
	D / s	45.5	42.7	43.2	41.6
	$S / times$	1.07	1.10	1.06	0.91

TABLE 8. Partial traffic data of Shenyang Station in Shenyang City (764766 detection points).

Time	Flow	Occupancy	Large vehicle flow	VMT	VHT	...	Delay
2020/8/3 8:00:00	656	21.6	5	115.6	3.6	..	0.2
2020/8/3 8:05:00	654	25.6	1	114.8	3.2	..	1.5
2020/8/3 8:10:00	657	25.9	35	114.1	3.1	..	2.3
...	...	...	...	...	...	..	...
2019/7/31 19:25:00	436	22.8	88	77.8	2.5	..	1.0
2019/7/31 19:30:00	478	16.7	101	82.4	2.3	..	0.7

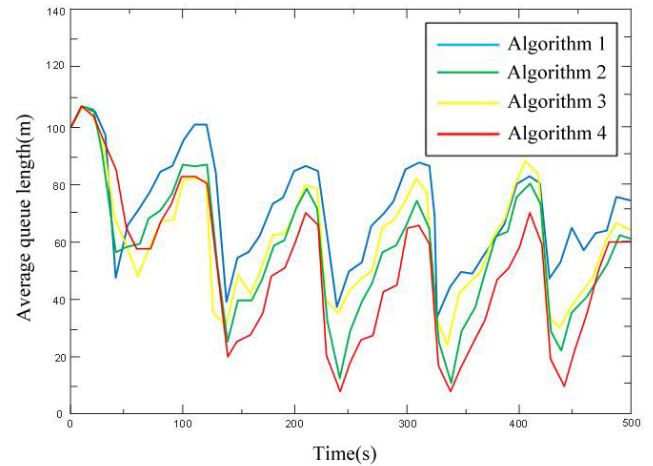
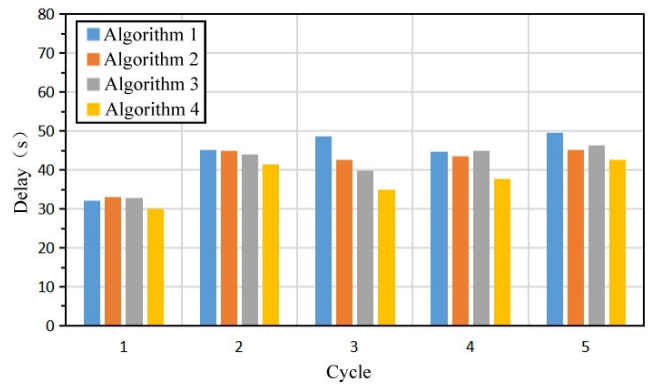
algorithm in this paper has the best control effect under various traffic congestion situations, and has certain advantages compared with other algorithms.

D. APPLICATION ANALYSIS OF TRAFFIC CONGESTION PREDICTION AND CONTROL BASED ON DIGITAL TWIN TECHNOLOGY

We adopt the traffic data obtained from 7 monitoring points on roads in some areas of Shenyang Station from August 1, 2020 to July 31, 2021. The distribution of monitoring points is shown in Figure 29, among which the monitoring point 722349 is the location to be predicted.

In the traffic state prediction experiment, 105,112 samples were collected, including 87,552 samples from August 1, 2020 to May 31, 2021, and 17,560 samples from June 1, 2020 to July 31, 2021. The partial traffic data of Shenyang Station are shown in Table 8.

Through the import of historical traffic data from some regional monitoring points of Shenyang Station, rapid modeling and management of traffic simulation were carried out. The data of vehicle number data, vehicle congestion period data and simulation target selection were respectively

**FIGURE 26.** Comparison of changes in average queue length.**FIGURE 27.** Comparison of changes in vehicle delays.

imported and set, and imported by intelligent algorithm. The traffic prediction model of IPSO-RBF and LSTM/SVM proposed in this paper, as well as the external algorithm decision tree, MLP, logistic regression, GBDT and random forest were imported, and the traffic state in this region was compared and predicted.

The proposed algorithm is compared with other algorithms in practical application through actual cases. The simulation environment for traffic state prediction is Intel (R) Core (TM) i5-5200U CPU @ 2.20GHz and MATLAB R2019a version. By time of day to predict how to forecast the traffic state, and to predict the result of comparing with the actual traffic status, red tag for the actual traffic status, the blue marker is forecast traffic status, traffic and divided into smooth, crowding and congestion degree of three different levels of state of traffic congestion, The comparison of traffic state prediction results of various comparison algorithms is shown in Figure 30 to Figure 35.

In Figure 30, the 24-hour traffic state is predicted according to the decision tree algorithm. In the actual state, the time is 7:45-9:00, and 12:45-13:15 is the congestion state and the congestion state respectively. However, the algorithm has missed the prediction state during this period of time,

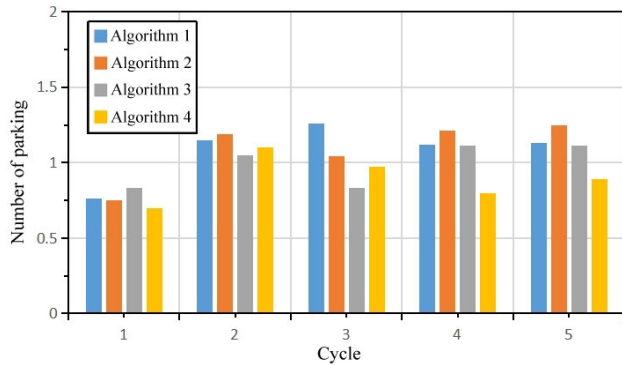


FIGURE 28. Schematic diagram of the distribution of some road monitoring points in Shenyang station.

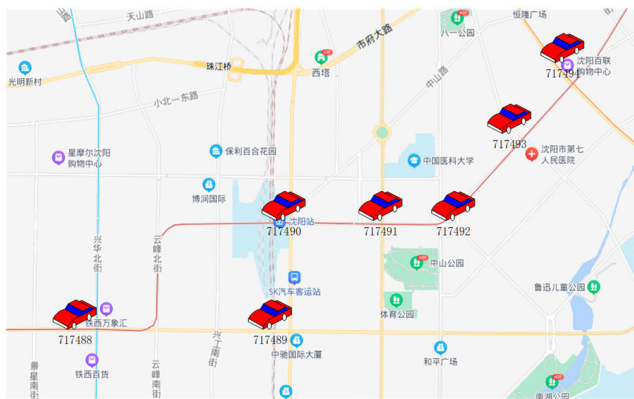


FIGURE 29. Comparison of changes in the number of stops.

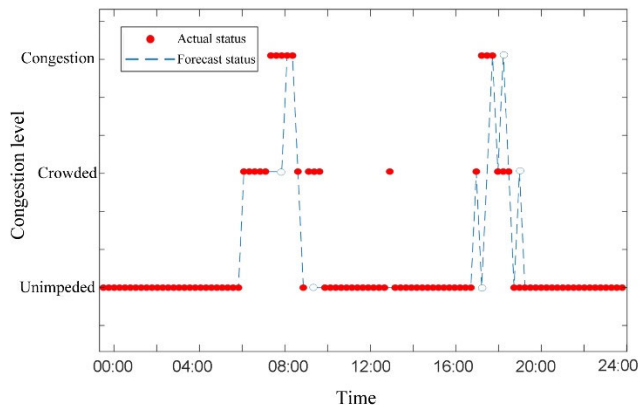


FIGURE 30. Decision tree prediction results.

resulting in the inconsistency between the traffic state during this period and the actual state; In the actual state, the traffic state is in congestion between 17:30-18:55, but the prediction result of this algorithm is congestion, and the congestion state will be displayed in the next phase. This algorithm also has some delay problems, which leads to the error of prediction results, so this algorithm has some prediction accuracy problems.

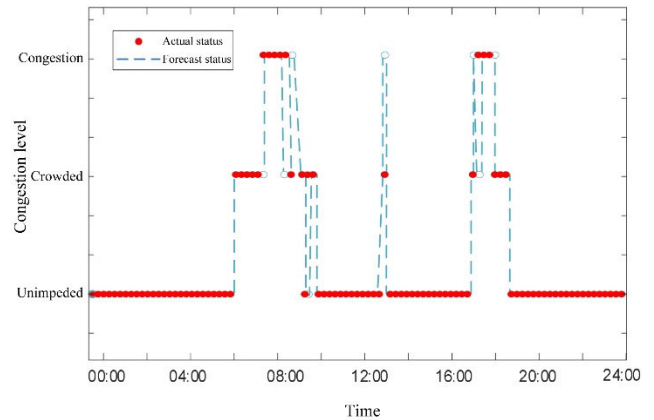


FIGURE 31. MLP prediction results.

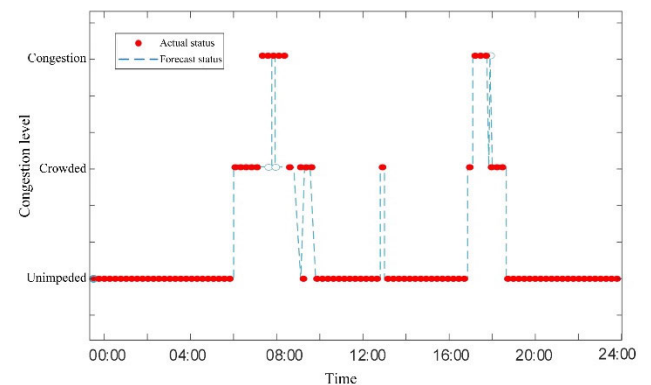


FIGURE 32. Logistic regression prediction results.

In Figure 31, the traffic status 24 hours a day is predicted according to MLP algorithm. In the actual status, the time is 7:30-7:45, and 8:45-9:00 is congestion status and congestion status respectively. However, the algorithm predicts that the traffic status during this period is congestion and congestion status, resulting in that the traffic status during this period is still in the actual traffic status of the previous stage; In the actual state, the traffic state is congested between 12:45-13:15, but the predicted result of the algorithm is congested, and the congestion state will be displayed in the next phase. The algorithm still has some problems with the prediction results, while the predicted state is congested at 17:30-18:30. In fact, the time period of the congestion state is smaller than the predicted time period. Therefore, the algorithm also has some problems with prediction accuracy.

In Figure 32, the 24-hour traffic state is predicted according to the logic regression algorithm. In the actual state, the time is 7:45-9:00, which is the congestion state. However, most of the time predicted by the algorithm during this period is the congestion state. Only around 8:00, the prediction result is the same as the actual prediction result, which leads to the feedback that the traffic state is mainly in the congestion state during this period, resulting in the wrong judgment of traffic grade, Impact on the next step of guidance and control

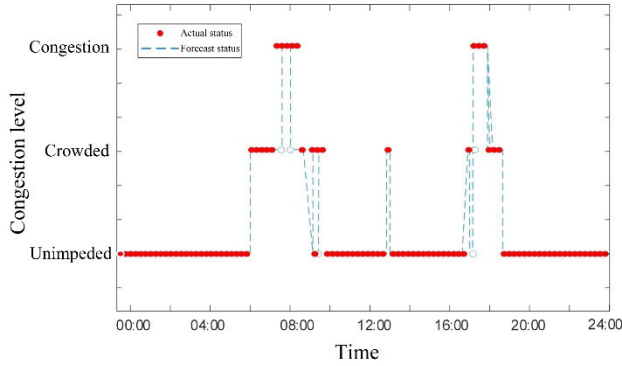


FIGURE 33. GBDT prediction results.

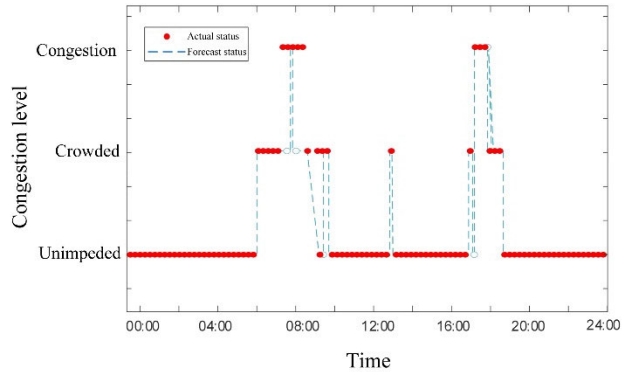


FIGURE 34. Random forest prediction results.

traffic congestion mitigation; However, the prediction results in the later time period are relatively accurate compared with the actual state. Therefore, this algorithm also has some prediction accuracy problems, and the prediction accuracy is much improved compared with the first two algorithms.

In Figure 33 and Figure 34, the traffic state of 24 hours a day is predicted according to GBDT and random forest algorithm. The results of the two algorithms and random forest algorithm are similar. In the actual state, the prediction results of the time period of 7:45-9:00 are congestion for most of the time, and the prediction results at around 8:00 are the same as the actual prediction results, leading to the feedback that the traffic state is mainly congestion during this period, It will also lead to wrong judgment of traffic grade, which will also affect the next step of guidance and control traffic congestion relief. Therefore, the two algorithms have some similarities with the logical regression algorithm in prediction, but there are also some problems in prediction accuracy.

In Figure 35, the traffic status 24 hours a day is predicted according to IPSO-RBF and LSTM/SVM models. Compared with the other four algorithms, the prediction results of the algorithm in this paper are obviously close to the actual traffic state in each time period. Therefore, the model in this paper has a certain accuracy in predicting the traffic state. Compared with other algorithms, it has certain advantages,

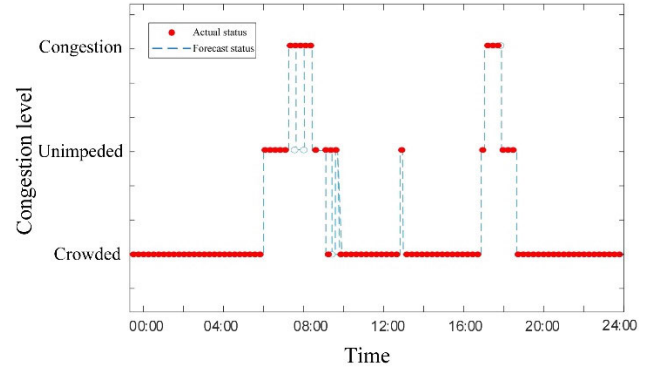


FIGURE 35. IPSO-RBF and LSTM/SVM prediction results.

and the feedback traffic state is more accurate, which is more conducive to making effective judgments and measures for the guidance and mitigation of traffic delay state.

According to classification prediction problem, need the forecast results are fair and just evaluation and comparative analysis, usually in the accuracy of the evaluation index as evaluation standard, but considering the actual sample label deviation problems, this paper by using precision rate and recall rate evaluation index is used to evaluate the result of the case analysis. Accuracy A , accuracy P and recall R are expressed as:

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (22)$$

$$P = \frac{TP}{TP + FP} \quad (23)$$

$$R = \frac{TP}{TP + FN} \quad (24)$$

In the formula: TP —Forecast positive value; FP —Predicted negative positive value; FN —Forecast positive negative value; TN —Negative forecast reverse.

The loss function is used for comparative analysis, which can be expressed as:

$$\text{Logloss} = -(y \log(p) + (1 - y) \log(1 - p)) \quad (25)$$

In the formula: y —Data real label; p —Probability of positive value predicted by algorithm.

Using cross entropy loss to solve various types of problems is expressed as:

$$\text{Multi-LogLoss} = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m y_{i,j} \log(p_{i,j}) \quad (26)$$

In the formula: n —Number of data; m —Data type; $y_{i,j}$ —True value; $p_{i,j}$ —Experimental result, i is the probability of type j .

The comparison between the prediction results of this model and other algorithms is shown in Table 9, where P represents the prediction accuracy of each algorithm; R represents the recall predicted by each algorithm; The delay state represents the sum of the road congestion state and the congestion state. The model in this paper has achieved good

TABLE 9. Comparison of prediction results of this model and other algorithms.

Model algorithm (Accuracy A)	Flow state		Crowded state		Congestion state		Delay state	
	<i>P</i>	<i>R</i>	<i>P</i>	<i>R</i>	<i>P</i>	<i>R</i>	<i>P</i>	<i>R</i>
Decision Tree=0.944	0.9	0.9	0.7	0.7	0.6	0.6	0.9	0.9
	77	64	64	70	13	01	33	01
MLP=0.952	0.9	0.9	0.8	0.8	0.6	0.6	0.9	0.9
	79	83	22	03	45	03	34	12
LR=0.953	0.9	0.9	0.7	0.8	0.6	0.5	0.9	0.9
	81	78	88	66	43	44	13	23
GBDT=0.955	0.9	0.9	0.8	0.8	0.7	0.5	0.9	0.9
	81	81	03	34	56	78	27	27
RF=0.956	0.9	0.9	0.8	0.8	0.7	0.6	0.9	0.9
	82	87	12	51	19	03	54	35
IPSO-RBF and LSTM/SVM=0.969	0.9	0.9	0.8	0.8	0.8	0.6	0.9	0.9
	91	93	44	83	67	39	64	48

prediction effect in each evaluation index, and has the best evaluation among all algorithms, which further verifies that the algorithm in this paper has high accuracy.

This section according to the operation characteristics of time and space, digital twin technology and traffic through this traffic state forecasting method and the use of other algorithms in Shenyang Station part of the traffic state forecasting comparison of traffic monitoring, on the basis of digital twin system, get a different algorithm corresponding to the traffic status of prediction results and the comparison of prediction accuracy. Among them, the predicted state of the IPSO-RBF and LSTM/SVM feature fusion model proposed in this paper is obviously close to the real traffic state, and the corresponding accuracy rate is 0.969, which is significantly higher than that of other algorithms. Therefore, it is further verified that the IPSO-RBF and LSTM/SVM feature fusion model under the digital twin technology has a better traffic state prediction effect on traffic congestion.

E. ANALYSIS OF TRAFFIC CONGESTION CONTROL RESULTS BASED ON VISSIM

For some road traffic experiment schemes of Shenyang Station in Shenyang City, we selected VISSIM simulation platform for simulation verification, and selected a part of one-way two-lane road section with a total length of about 3km for simulation experiment. According to the traffic data monitored by the road section, we set the actual traffic flow, speed and other relevant traffic flow parameters, and analyzed the results of the road section.

Based on the data of the congested road section obtained from the digital twin traffic congestion prediction and control system platform, combined with the congestion road section control method proposed above based on flow distribution, the simulation analysis of different road sections is carried out, and finally the analysis process and results of traffic congestion diffusion speed in different upstream and downstream areas are obtained, as shown in Table 10 to Table 12.

TABLE 10. Congestion diffusion speed of two-lane roads based on traffic congestion control area.

Time	Upstream Region (Zone A)			Downstream Region (Zone B)			Congestion diffusion rate (Absolute value)
	Speed	Flow	Density	Speed	Flow	Density	
	Speed	Flow	Density	Speed	Flow	Density	
1	57.0	716	9.03	55.1	656	8.56	0
2	57.4	843	10.4	57.3	836	8.13	0
3	57.6	989	8.76	57.2	935	10.26	0
4	56.8	1126	10.23	54.1	1014	14.23	0
5	55.0	1123	11.53	49.6	1014	22.36	1.35
6	54.7	1011	13.43	49.9	1035	23.56	3.34
7	51.1	1033	18.75	14.3	975	66.08	4.23
8	34.5	1109	36.44	14.6	986	64.33	4.69
9	31.6	1103	41.23	13.4	840	65.12	5.76
10	21.3	1036	57.23	14.2	869	67.11	5.89
11	17.8	1065	48.98	13.7	937	67.23	7.11
12	18.3	1036	48.11	11.4	883	72.12	10.22
13	18.6	1134	57.56	10.3	864	75.36	8.91
14	18.4	1156	57.64	10.4	884	75.34	13.45
15	18.3	1145	53.01	9.8	870	77.23	13.45
16	18.1	1236	51.40	13.1	883	75.12	15.46
17	19.3	1145	52.10	11.3	863	76.56	10.36
18	19.4	1038	51.23	10.7	877	77.46	8.45
19	19.6	1010	51.11	12.3	883	73.12	6.12
20	20.1	986	58.44	10.4	883	77.66	2.24
21	20.3	924	57.66	11.5	876	77.36	1.23
22	20.7	862	55.78	10.3	851	76.28	0.86
23	21.6	893	54.33	14.1	850	77.26	0.63
24	22.1	881	54.31	10.1	843	74.14	0.42

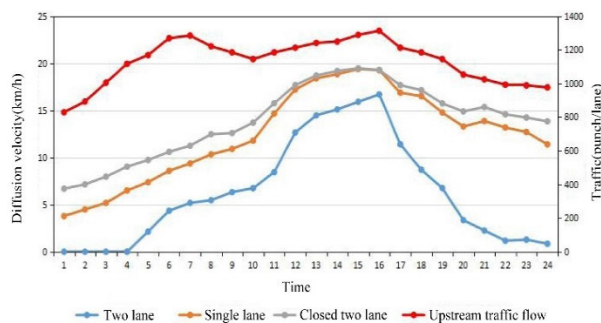
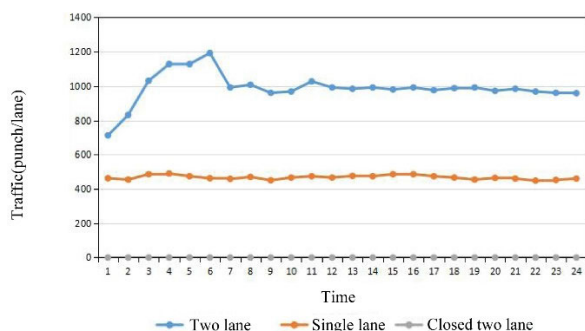
TABLE 11. Congestion diffusion speed of single lane based on traffic congestion control area.

Time	Upstream Region (Zone A)			Downstream Region (Zone B)			Congestion diffusion rate (Absolute value)
	Speed	Flow	Density	Speed	Flow	Density	
	Speed	Flow	Density	Speed	Flow	Density	
1	57.0	716	9.03	4.5	351	99.33	2.66
2	57.4	843	10.4	4.6	343	98.32	3.37
3	57.6	989	8.76	4.3	345	99.16	4.06
4	56.8	1126	10.23	4.6	380	98.63	5.38
5	55.0	1123	11.53	4.3	363	100.65	6.27
6	54.7	1011	13.43	4.2	351	100.36	7.46
7	51.1	1033	18.75	4.3	347	100.01	8.28
8	34.5	1109	36.44	4.2	360	100.60	9.13
9	31.6	1103	41.23	4.5	340	100.23	9.86
10	21.3	1036	57.23	4.3	355	100.78	10.67
11	17.8	1065	48.98	4.5	363	100.19	13.56
12	18.3	1036	48.11	4.1	355	100.56	16.01
13	18.6	1134	57.56	4.0	365	100.78	17.78
14	18.4	1156	57.64	3.8	352	105.43	17.56
15	18.3	1145	53.01	4.0	375	105.64	18.27
16	18.1	1236	51.40	4.1	373	100.45	18.15
17	19.3	1145	52.10	4.0	363	105.03	15.75
18	19.4	1038	51.23	4.2	355	105.36	15.45
19	19.6	1010	51.11	4.3	343	101.63	13.65
20	20.1	986	58.44	4.4	353	100.98	12.16
21	20.3	924	57.66	4.3	350	101.13	12.75
22	20.7	862	55.78	4.5	337	101.89	12.06
23	21.6	893	54.33	4.6	341	100.39	11.58
24	22.1	881	54.31	4.1	350	101.42	10.27

Through the above calculation of vehicle speed, flow and density in different upstream and downstream areas, the corresponding congestion diffusion speed results are obtained, the results are analyzed, and the changes of vehicle flow and congestion diffusion speed in the upstream area and the changes of vehicle flow in the downstream congestion control area are obtained, as shown in Figure 36 and Figure 37.

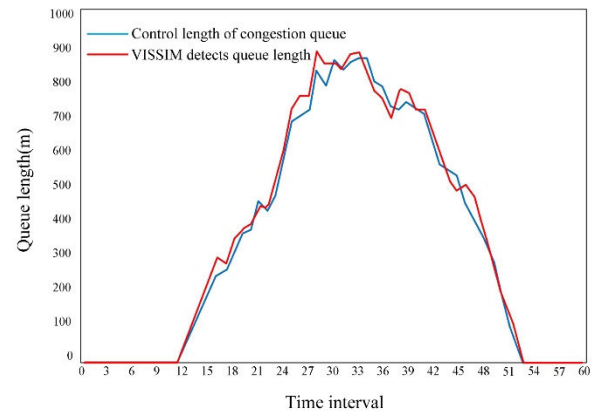
TABLE 12. Congestion diffusion speed based on closed two-lane road in traffic congestion control area.

Time	Upstream Region (Zone A)			Downstream (Zone B)		Region	Congestion diffusion rate (Absolute value)
	Speed	Flow	Density	Speed	Flow		
1	57.0	716	9.03	0	0	127.00	5.57
2	57.4	843	10.4	0	0	127.00	6.03
3	57.6	989	8.76	0	0	127.00	6.85
4	56.8	1126	10.23	0	0	127.00	8.01
5	55.0	1123	11.53	0	0	127.00	8.71
6	54.7	1011	13.43	0	0	127.00	9.48
7	51.1	1033	18.75	0	0	127.00	10.13
8	34.5	1109	36.44	0	0	127.00	11.35
9	31.6	1103	41.23	0	0	127.00	11.26
10	21.3	1036	57.23	0	0	127.00	12.56
11	17.8	1065	48.98	0	0	127.00	13.65
12	18.3	1036	48.11	0	0	127.00	16.56
13	18.6	1134	57.56	0	0	127.00	17.56
14	18.4	1156	57.64	0	0	127.00	18.06
15	18.3	1145	53.01	0	0	127.00	18.35
16	18.1	1236	51.40	0	0	127.00	18.14
17	19.3	1145	52.10	0	0	127.00	16.54
18	19.4	1038	51.23	0	0	127.00	16.03
19	19.6	1010	51.11	0	0	127.00	14.62
20	20.1	986	58.44	0	0	127.00	13.77
21	20.3	924	57.66	0	0	127.00	14.24
22	20.7	862	55.78	0	0	127.00	13.46
23	21.6	893	54.33	0	0	127.00	13.12
24	22.1	881	54.31	0	0	127.00	13.75

**FIGURE 36.** Changes in traffic flow and congestion diffusion speed in the upstream area.**FIGURE 37.** Traffic flow changes in the downstream congestion control area.

According to Figure 36 and Figure 37:

(1) With the change of the adjustment of the number of lanes in congested sections, that is, the distribution of traffic flow in congested sections, the change trend of the congestion

**FIGURE 38.** VISSIM simulation detection and control congestion queue length difference comparison chart.

diffusion speed is roughly the same as that of the upstream traffic flow. It shows that the diffusion speed of traffic congestion is proportional to the change of traffic flow.

(2) The degree of traffic congestion is inversely proportional to the speed of congestion diffusion. In the case of two-lane traffic flow, the speed of congestion diffusion fluctuates greatly. In the late stage of traffic congestion, the speed of congestion diffusion decreases greatly, while in the case of single lane or closed two lane traffic, the amplitude of fluctuation of angular diffusion is roughly similar and the amplitude of fluctuation is flat.

(3) In consideration of the potential congestion in the future, the safety queuing coefficient of vehicles is used to reasonably use the road occupation in advance, effectively reducing the speed of congestion diffusion.

(4) Without considering the potential congested road sections in the future, the traffic volume will decrease significantly as the congestion time continues to increase.

It can be seen that the result analysis is consistent with the algorithm research in this paper, and the congestion diffusion speed is closely related to the reasonable traffic flow and the consideration of future potential road allocation.

According to the safety queuing coefficient proposed above, the congestion queuing length is controlled. The congestion queuing length is compared with the measured value in the VISSIM simulation platform, as shown in Figure 38. The absolute and relative error values of the queuing length of the two methods at each time point are calculated and analyzed, and the results are shown in Figure 39.

It can be seen from Figure 38 that the control method proposed in this paper effectively controls the queue length of vehicles. The maximum queue length of road vehicles is 875.6m. The result after using the control method in this paper is 860.4m, with a difference of 15.2m and a relative error of 1.73%. It is enough to show that the degree of road congestion can be effectively reduced by the control method in this paper.

It can be seen from Figure 39 that from the formation to the end of road congestion, the absolute error is 66.89m, and

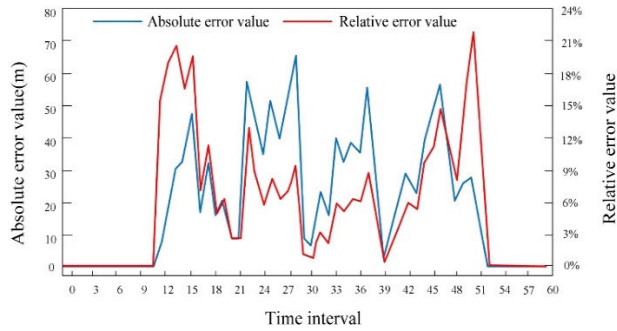


FIGURE 39. VISSIM simulation detection and control congestion queue length error analysis.

the relative error is 18.33%; The average absolute error is 19.97m, and the average relative error is 4.73%, which further shows that the algorithm in this paper has a good control effect when controlling congested roads.

IV. CONCLUSION

Aiming at the problem of traffic congestion prediction and control in the field of intelligent transportation, this paper studies traffic state prediction and congestion control by combining the theory of traffic prediction and control and the theory of intelligent algorithm. Firstly, the factors that affect the traffic state are determined by the use of traffic related theory knowledge. Secondly, aiming at the problem of traffic state prediction, a traffic state prediction model based on IPSO-RBF and LSTM/SVM feature fusion is proposed. Thirdly, aiming at the traffic congestion problem, a traffic congestion control method based on traffic allocation is proposed and a case study is carried out. The main research work and contributions of this paper are as follows:

(1) A traffic state prediction model based on IPSO-RBF and LSTM/SVM feature fusion is proposed. Firstly, according to the characteristics of traffic operation, the general flow framework of traffic state prediction process is established; Secondly, preprocess the traffic operation data monitored by the monitor, set the criteria for evaluating abnormal data, and use the box chart to filter and repair the abnormal data; Then, combined with the time-space characteristics of traffic operation, IPSO-RBF and LSTM feature extraction models are used to extract the time-space and time series of traffic data respectively, and the extracted features are fused and input into SVM to predict the traffic status; Finally, through the acquisition of traffic sample data in some regions of Shenyang, a comparison experiment was designed, and other models were used to compare the fitting values and prediction results with the model in this paper. In the error comparison analysis of the model in this paper, *RMSE* was 13.2653, *MAPE* was 13.6345, and *MAE* was 11.0113, *AIC* was -5.991 and *BIC* was -7.511 , which were the lowest compared with other models. The results showed the feasibility and accuracy of the algorithm in this paper.

(2) A congestion control method based on traffic assignment is proposed. Firstly, according to the characteristics of traffic congestion, the factors affecting traffic congestion are analyzed; Then, using the methods of road capacity judgment and vehicle safety queuing coefficient, the residual capacity of the road is calculated, and the future road sections with congestion are reasonably allocated with traffic flow, so as to effectively guide the congested road sections. Finally, the simulation modeling experiment based on VISSIM shows the superiority of the congestion control method of traffic allocation in this paper.

REFERENCES

- [1] W. Pan, Y. Xue, H.-D. He, and W.-Z. Lu, "Impacts of traffic congestion on fuel rate, dissipation and particle emission in a single lane based on NASCH model," *Phys. A, Stat. Mech. Appl.*, vol. 503, pp. 154–162, Aug. 2018.
- [2] X. Luo, D. Li, Y. Yang, and S. Zhang, "Spatiotemporal traffic flow prediction with KNN and LSTM," *J. Adv. Transp.*, vol. 2019, pp. 1–10, Feb. 2019.
- [3] F. Zhao, G.-Q. Zeng, and K.-D. Lu, "EnLSTM-WPEO: Short-term traffic flow prediction by ensemble LSTM, NNCT weight integration, and population extremal optimization," *IEEE Trans. Veh. Technol.*, vol. 69, no. 1, pp. 101–113, Jan. 2020.
- [4] V. Osipov, V. Nikiforov, N. Zhukova, and D. Miloserdov, "Urban traffic flows forecasting by recurrent neural networks with spiral structures of layers," *Neural Comput. Appl.*, vol. 32, no. 18, pp. 14885–14897, 2020.
- [5] P. He, G. Jiang, S.-K. Lam, and Y. Sun, "Learning heterogeneous traffic patterns for travel time prediction of bus journeys," *Inf. Sci.*, vol. 512, pp. 1394–1406, Feb. 2020.
- [6] S. Guo, Y. Lin, S. Li, Z. Chen, and H. Wan, "Deep spatial-temporal 3D convolutional neural networks for traffic data forecasting," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 10, pp. 3913–3926, Oct. 2019.
- [7] J. Zhu, C. Huang, M. Yang, and G. P. Cheong Fung, "Context-based prediction for road traffic state using trajectory pattern mining and recurrent convolutional neural networks," *Inf. Sci.*, vol. 473, pp. 190–201, Jan. 2019.
- [8] Y. Lin, X. Dai, L. Li, and F.-Y. Wang, "Pattern sensitive prediction of traffic flow based on generative adversarial framework," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 6, pp. 2395–2400, Jun. 2018.
- [9] H. Peng, H. Wang, B. Du, M. Z. A. Bhuiyan, H. Ma, J. Liu, L. Wang, Z. Yang, L. Du, S. Wang, and P. S. Yu, "Spatial temporal incidence dynamic graph neural networks for traffic flow forecasting," *Inf. Sci.*, vol. 521, pp. 277–290, Jun. 2020.
- [10] L. Zhao, Y. Zhou, H. Lu, and H. Fujita, "Parallel computing method of deep belief networks and its application to traffic flow prediction," *Knowl.-Based Syst.*, vol. 163, pp. 972–987, Jan. 2019.
- [11] L. Li, L. Qin, X. Qu, J. Zhang, Y. Wang, and B. Ran, "Day-ahead traffic flow forecasting based on a deep belief network optimized by the multi-objective particle swarm algorithm," *Knowl.-Based Syst.*, vol. 172, pp. 1–14, May 2019.
- [12] J. Guo, Z. Liu, W. Huang, Y. Wei, and J. Cao, "Short-term traffic flow prediction using fuzzy information granulation approach under different time intervals," *IET Intell. Transp. Syst.*, vol. 12, no. 2, pp. 143–150, Mar. 2018.
- [13] U. Ryu, J. Wang, T. Kim, S. Kwak, and U. Juhyok, "Construction of traffic state vector using mutual information for short-term traffic flow prediction," *Transp. Res. C, Emerg. Technol.*, vol. 96, pp. 55–71, Nov. 2018.
- [14] S. O. M. Kashi and M. Akbarzadeh, "A framework for short-term traffic flow forecasting using the combination of wavelet transformation and artificial neural networks," *J. Intell. Transp. Syst.*, vol. 23, no. 1, pp. 60–71, 2019.
- [15] C. Luo, C. Huang, J. Cao, J. Lu, W. Huang, J. Guo, and Y. Wei, "Short-term traffic flow prediction based on least square support vector machine with hybrid optimization algorithm," *Neural Process. Lett.*, vol. 50, pp. 2305–2322, Mar. 2019.
- [16] X. Feng, X. Ling, H. Zheng, Z. Chen, and Y. Xu, "Adaptive multi-kernel SVM with spatial-temporal correlation for short-term traffic flow prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 6, pp. 2001–2013, Jun. 2019.

- [17] P. Duan, G. Mao, W. Liang, and D. Zhang, "A unified spatio-temporal model for short-term traffic flow prediction," *IEEE Trans. Intel. Transp. Syst.*, vol. 20, no. 9, pp. 3212–3223, Sep. 2019.
- [18] G. Dai, C. Ma, and X. Xu, "Short-term traffic flow prediction method for urban road sections based on space-time analysis and GRU," *IEEE Access*, vol. 7, pp. 143025–143035, 2019.
- [19] S. Duan, W. Song, E. Zio, C. Cattani, and M. Li, "Product technical life prediction based on multi-modes and fractional Lévy stable motion," *Mech. Syst. Signal Process.*, vol. 161, Dec. 2021, Art. no. 107974.
- [20] W. Song, H. Liu, and E. Zio, "Long-range dependence and heavy tail characteristics for remaining useful life prediction in rolling bearing degradation," *Appl. Math. Model.*, vol. 102, pp. 268–284, Feb. 2022.
- [21] H. Liu, W. Song, Y. Zhang, and A. Kudreyko, "Generalized Cauchy degradation model with long-range dependence and maximum Lyapunov exponent for remaining useful life," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.
- [22] X. Yang, Y. Zou, and L. Chen, "Operation analysis of freeway mixed traffic flow based on catch-up coordination platoon," *Accident Anal. Prevention*, vol. 175, Sep. 2022, Art. no. 106780.
- [23] M. Saleem, S. Abbas, T. M. Ghazal, M. Adnan Khan, N. Sahawneh, and M. Ahmad, "Smart cities: Fusion-based intelligent traffic congestion control system for vehicular networks using machine learning techniques," *Egyptian Informat. J.*, vol. 23, no. 3, pp. 417–426, Sep. 2022.
- [24] A. Atta, S. Abbas, M. A. Khan, G. Ahmed, and U. Farooq, "An adaptive approach: Smart traffic congestion control system," *J. King Saud Univ., Comput. Inf. Sci.*, vol. 32, no. 9, pp. 1012–1019, Nov. 2020.



JIANAN XU was born in 1997. He is currently pursuing the master's degree with Shenyang University. His research interest includes state recognition and prediction.



SIRUI CONG was born in 1998. She is currently pursuing the master's degree with Shenyang University. Her main research interests include data processing and state prediction.



CHUANSHENG QU was born in 1996. He received the master's degree from Shenyang University. His main research interest includes traffic recognition and prediction.



TIANRUI ZHANG was born in Shenzhou, Hebei, China, in 1985. He received the Ph.D. degree from Northeastern University. He is currently an Associate Professor with Shenyang University. His research interests include state prediction and system optimization.



WEIBO ZHAO was born in 2000. He is currently pursuing the degree with Shenyang University.

• • •